Approach 1: Latent Self-Correction, From Detection to Correction

A complete account of the Approach 1 work on Qwen3-VL-2B: what was built, why each step was taken, and what every experiment showed. Written for the mid-point report and presentation.

Model: Qwen3-VL-2B (frozen). Datasets: ChartQA, TextVQA, DocVQA, ScienceQA. Every metric is on held-out test splits unless stated. Detailed per-experiment reports are linked in the text and listed in Appendix A.

0.719 to 0.881
Pooled AUROC, answer-token probability to unified head

0.750 to 0.895
ChartQA AUROC, original study, probability to linear probe

+0.13 to +0.27
AUROC gain of the learned heads on every dataset

0.011
AUROC forgotten by the unified head over four datasets

0.92 to 0.99
Share of correct answers whose score falls under noise, old to new loss

540
Trained heads in the repository, 2,049 to 2.1 million parameters

1. The goal and the road map

Approach 1 asks a vision-language model to notice when its own visual evidence is weak and to act on that before answering. It has two halves.

Detection (the gate). A small network reads the hidden state of the frozen model and outputs the probability that the answer is correct. If this works, the model can be told when to slow down, retry, or hand the question to something else.

Correction. A step that acts on answers the gate flags. Everything in Sections 3 to 11 builds and tests the gate. Section 12 reports the first experiments on acting on it.

Where the correctness head sits



The language model is frozen throughout. Only the small head is trained. The correction stage on the right is where the experiments of Section 12 apply.

1.1 The ten stages, in the order they happened

#	Stage	What was done and why	Main result	Report
1	Baseline: answer-token probability	See whether the model's own confidence can tell right answers from wrong ones	AUROC 0.750 on ChartQA and 0.719 pooled. It equals 1.0 on 77% to 90% of answers	01, 02
2	Original ChartQA study	Train small heads on frozen hidden states: a pre-generation head and a verifier	AUROC 0.89, error AUPRC 0.74, 97% accuracy on the most trusted half	01
3	Three more datasets	Check the result is not a ChartQA artifact: TextVQA, DocVQA, ScienceQA	Heads beat the answer-token probability by 0.13 to 0.27 AUROC on every dataset	02
4	Unified head	One network for both jobs: score before the answer and after it	Matches or beats the separate heads on every dataset	02
5	Pooled mixture	Train and test on all four datasets at once, old loss	Unified head AUROC 0.881 against 0.719 for the answer-token probability	05
6	New loss, swapped image	Add a visual-grounding term so the score falls when the image is wrong	Grounding rises from 93% to 96% with no loss of AUROC	05
7	New loss, noised image	Same loss, but corrupt the image by adding noise instead of swapping it	Grounding rises from 92% to 99% and carries over to other corruptions	06
8	Sequential training	Train on A, then B, then C, and check A is still remembered	Forgetting 0.011 to 0.046 AUROC depending on head and loss; replay removes most of it for the verifier	05, 06
9	Trained heads exported	Save every trained head so teammates can build on it	540 heads, one to eight megabytes each	this report
10	Correction experiments	Use the head to pick or revise answers (retry, best-of-five, deeper pass)	Head selection among five samples: +0.4 points over greedy on the full test split (not significant); the head ranks candidates well (AUROC 0.883 against 0.815)	07

Report numbers refer to the files in `results/approach1/`: 01 original ChartQA study, 02 multi-dataset report, 03 metric tables, 05 mixture and new loss, 06 noise corruption, 07 correction experiments.

> **TERMS USED THROUGHOUT**
>
> The **answer-token probability** is the untrained baseline: the probability the model itself gave its answer tokens. A **linear probe** is a trained head with no hidden layer. A **head** is any small trained network that reads the frozen hidden states. **AUROC** is the chance that a randomly chosen correct answer gets a higher score than a randomly chosen wrong one.

2. Foundations: what is measured and how

2.1 The model and what is cached

The vision-language model is Qwen3-VL-2B, kept frozen for the whole project. It answers every question once. For each question the last-layer hidden state of the last token is cached in three situations, so that all later experiments reuse them and never run the large model again.

- **Prompt state** h_q: the image and the question, ending at the generation prompt. It exists before any answer does.
- **Answer state** h_a: the image, the question, and the model's own answer appended. It describes a specific candidate answer.
- **Reference state** h_{ref}: the same, with the correct reference answer appended. Used only as training supervision.

Each state is a vector of size $d = 2048$. Caching is what makes the experiments cheap. Extracting the states of one dataset split takes up to an hour and a half on one GPU (82 minutes for the 28,299 ChartQA training questions with a corrupted image), but a whole pooled sweep of head fits on the cached states takes 3 to 12 minutes per head kind. The price is that nothing the head learns can change what the model sees or says.

2.2 What counts as a correct answer

Each dataset has its own rule, converted to one binary label $y \in \{0, 1\}$. Correctness is judged by the dataset's official scoring, not by exact string match, because exact match would call a right answer wrong whenever formatting differs.

Dataset	Train	Validation	Test	Base accuracy, train	Base accuracy, test	Correct means
ChartQA	28,299	1,920	2,500	78.5%	75.8%	Relaxed accuracy (5% for numbers)
TextVQA	8,000	1,500	2,500	82.3%	81.0%	VQA soft accuracy at least 0.5
DocVQA	8,000	1,500	2,500	93.2%	87.0%	ANLS at least 0.9
ScienceQA	6,218	1,500	2,017	86.5%	86.7%	Option letter matches

ChartQA uses relaxed accuracy (a numeric answer within 5% of the reference). TextVQA uses the VQA soft accuracy over ten human answers, correct at 0.5 or more. DocVQA uses ANLS, correct at 0.9 or more. ScienceQA uses the option letter. Splits are grouped by image or document so that no image appears in two splits, and the validation and test splits of TextVQA and DocVQA are carved out of the official validation split because the official test answers are hidden.

2.3 Metrics

The head outputs a score s and the probability $p = \sigma(s/T)$, where σ is the logistic function and T is a temperature fitted once on the validation split by minimizing the cross-entropy (temperature scaling, Guo et al.

2017). The test split is used only for the final numbers.

- **Correctness AUROC**, $\mathrm{AUROC} = P(s_{\text{correct}} > s_{\text{wrong}})$, the probability that a random correct answer outranks a random wrong one. 0.5 is a coin flip, 1.0 is perfect.
- **Error AUPRC**, the area under the precision-recall curve when the *wrong* answers are the positive class. It measures how well errors are packed at the top of a queue sorted from least to most trusted, which is the view that matters for a gate. Chance equals the error rate.
- **Brier score**, $\frac{1}{n}\sum_i (p_i - y_i)^2$. Lower is better. It rewards being right and being honest about uncertainty.
- **ECE**, the expected calibration error, $\sum_b \frac{|B_b|}{n}\left|\mathrm{acc}(B_b) - \mathrm{conf}(B_b)\right|$ over ten probability bins. Lower is better.
- **Accuracy at coverage**: keep the most trusted fraction of answers and report their accuracy. Full coverage equals the base accuracy.
- **Errors captured at a flag rate**: flag the lowest-scored fraction of answers and report the share of all errors that fall inside.

2.4 Selection protocol

Every configuration is trained on three seeds. A winner is the configuration with the best mean validation error AUPRC (validation AUROC breaks ties), chosen on the validation split only. Test numbers are read afterwards for the winners. Where a number is shown with a plus-minus sign, it is the standard deviation across seeds. This matters because several differences in this project are a few thousandths of AUROC, which is the size of the seed noise.

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3. Stage 1: the model's own confidence

The natural first idea is to use what the model already says about itself. When Qwen3-VL generates an answer of m tokens, it assigns each token a probability. The geometric mean of those probabilities is a confidence for the whole answer:

$$c = \exp\Big(\frac{1}{m}\sum_{j=1}^{m} \log p(t_j \mid t_{<j},\ \text{image},\ \text{question})\Big)$$

It needs no training and no extra computation. We call it the **answer-token probability** throughout. Its problem is saturation: modern instruction-tuned models are very sure of their tokens whether or not the answer is right, so the score is exactly 1.0 for most answers and cannot rank them.

Dataset	Base accuracy	Token probability exactly 1.0	AUROC	Error AUPRC	Brier	ECE
ChartQA	75.8%	76.6%	0.751	0.581	0.180	0.175
TextVQA	81.0%	79.0%	0.726	0.449	0.158	0.152
DocVQA	87.0%	82.4%	0.685	0.333	0.114	0.106
ScienceQA	86.7%	89.6%	0.654	0.323	0.116	0.108

Dataset	Base accuracy	Token probability exactly 1.0	AUROC	Error AUPRC	Brier	ECE
Pooled mixture	82.4%	81.5%	0.719	0.443	0.143	0.136

> **WHY THIS STAGE MATTERS**
> On the original ChartQA study the score was exactly 1.0 for 1,921 of 2,500 test answers (76.8%), including many wrong ones. Keeping its most confident half of answers raised accuracy only from 75.8% to 92.4%. Across the four datasets the score sits at 1.0 for 77% to 90% of answers, so a threshold on it cannot separate right from wrong. This is the gap the learned heads are built to close, and it is the comparison every later result is measured against.

An answer-length baseline (does a longer answer tend to be wrong?) was also computed as a sanity check on shortcuts. It reaches an AUROC of only 0.52 on the pooled mixture, so the heads below are not simply learning answer length.

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4. Stage 2: the original ChartQA study

Question. Before building anything that corrects a model, can the model's internal state tell us when it is about to be wrong? This was first tested on ChartQA alone. The full study is report 01.

4.1 Two heads, two questions

- The **pre-generation head** reads h_q and predicts whether the question will be answered correctly before any answer exists. It measures how hard the question looks to the model.
- The **verifier head** reads h_a and judges one specific generated answer. This is the head a correction step needs, because it scores the candidate itself.

Both are small networks trained on a correct or wrong label, with the language model frozen. The reason for two heads is that they answer different questions at different times: one before the answer (useful to decide whether to generate more carefully), one after (useful to decide whether to keep or revise).

4.2 Architecture and training

The default head is a two-layer network: layer normalization, a linear layer to 512 units, GELU, 10% dropout, one output. It was trained with AdamW (learning rate 0.001, weight decay 0.0001, batch size 512, 20 epochs). The later sweep also tested a plain **linear probe** $s = w^\top h + b$ and a 256-unit variant. The parameter counts are in Section 5.

4.3 Loss functions of the original study

Both heads use binary cross-entropy plus a ranking loss. The verifier adds a paired loss that uses the reference answer.

$$L_{\text{BCE}} = -\big[y \log \sigma(s) + (1-y)\log(1-\sigma(s))\big]$$

$$L_{\text{rank}} = \underset{i \in \text{correct},\, j \in \text{wrong}}{\text{mean}} \text{softplus}\big(-(s_i - s_j)\big), \qquad \text{softplus}(z) = \log(1+e^z)$$

$$L_{\text{pair}} = \underset{j \in \text{wrong}}{\text{mean}} \text{softplus}\big(-(s_{\text{ref},j} - s_{\text{cand},j})\big)$$

$$L_{\text{pre}} = L_{\text{BCE}} + 0.25\, L_{\text{rank}}, \qquad L_{\text{verifier}} = L_{\text{BCE}} + 0.25\, L_{\text{rank}} + 0.25\, L_{\text{pair}}$$

The ranking term compares every correct example with every wrong one in a batch. The paired term compares the model's wrong answer with the correct reference answer for the same image and question, using two teacher-forced passes that differ only in the answer text. The reference is training supervision only: at deployment the verifier sees the image, question and candidate, never the unknown correct answer. The idea was to teach the verifier what a right answer looks like by contrast.

4.4 The sweep

Using cached features, 12 configurations per head, each on three seeds (36 runs per head, 72 in total), were trained: architecture ∈ {linear, MLP-256, MLP-512}, learning rate ∈ {3e-4, 1e-3}, ranking weight ∈ {0, 0.25}, seed ∈ {1, 2, 3}. A second sweep added early stopping and five ranking weights for the verifier. The table gives mean validation AUROC of the 72 runs, grouped. The raw runs are in `04_sweep_data/chartqa_original_validation_runs.json`.

Head	Architecture	Ranking weight	Parameters	Mean validation AUROC	Range over runs
Pre-generation	linear	0	2,049	0.8407	0.8362 to 0.8457
Pre-generation	linear	0.25	2,049	0.8390	0.8333 to 0.8446
Pre-generation	mlp_256	0	528,897	0.8278	0.8153 to 0.8363
Pre-generation	mlp_256	0.25	528,897	0.8260	0.8108 to 0.8364
Pre-generation	mlp_512	0	1,053,697	0.8235	0.8130 to 0.8351
Pre-generation	mlp_512	0.25	1,053,697	0.8226	0.8108 to 0.8332
Verifier	linear	0	2,049	0.8349	0.8308 to 0.8394
Verifier	linear	0.25	2,049	0.8154	0.8132 to 0.8170
Verifier	mlp_256	0	528,897	0.8407	0.8373 to 0.8435
Verifier	mlp_256	0.25	528,897	0.8228	0.8187 to 0.8279
Verifier	mlp_512	0	1,053,697	0.8419	0.8390 to 0.8435
Verifier	mlp_512	0.25	1,053,697	0.8218	0.8160 to 0.8270

4.5 Results (test split, 2,500 questions)

Metric	Answer-token probability	Pre-gen MLP	Pre-gen linear probe	Verifier MLP (swept)	Verifier, early stop (3 seeds)
Correctness AUROC	0.750	0.863	0.895	0.887	0.891 ± 0.001
Error AUPRC	0.583	0.669	0.732	0.723	0.738 ± 0.005
Brier score	0.178	0.123	0.109	0.112	0.111 ± 0.001
ECE (10 bins)	0.172	0.055	0.036	0.030	0.038 ± 0.014
Accuracy at 50% coverage	92.40%	95.12%	97.04%	96.64%	97.04%

> **WHAT THE STUDY FOUND**
> (1) The answer-token probability is weak (AUROC 0.750). (2) A small head on frozen states is far better: AUROC 0.89, error AUPRC about 0.74, Brier score 0.178 to 0.11, ECE 0.172 to between 0.03 and 0.04, and accuracy on the most trusted half of answers from 92.4% to about 97%. (3) The linear probe had the highest AUROC (0.895), which says expected difficulty is close to linearly readable from the question state. The early-stopped verifier had the highest error AUPRC (0.738) and is the relevant model for judging a specific answer. (4) The ranking objectives did not help: plain cross-entropy won both heads in the main sweep.

Why these choices were made. Linear probes were included because a linear probe that works is the strongest evidence that the information is already in the hidden state, and not created by the head. The sweep was run on cached features so that the choice of architecture, learning rate and ranking weight rests on 72 fits and not on one lucky run. Selection used validation error AUPRC because for a gate, finding the wrong answers matters more than the average ranking.

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5. Head sizes

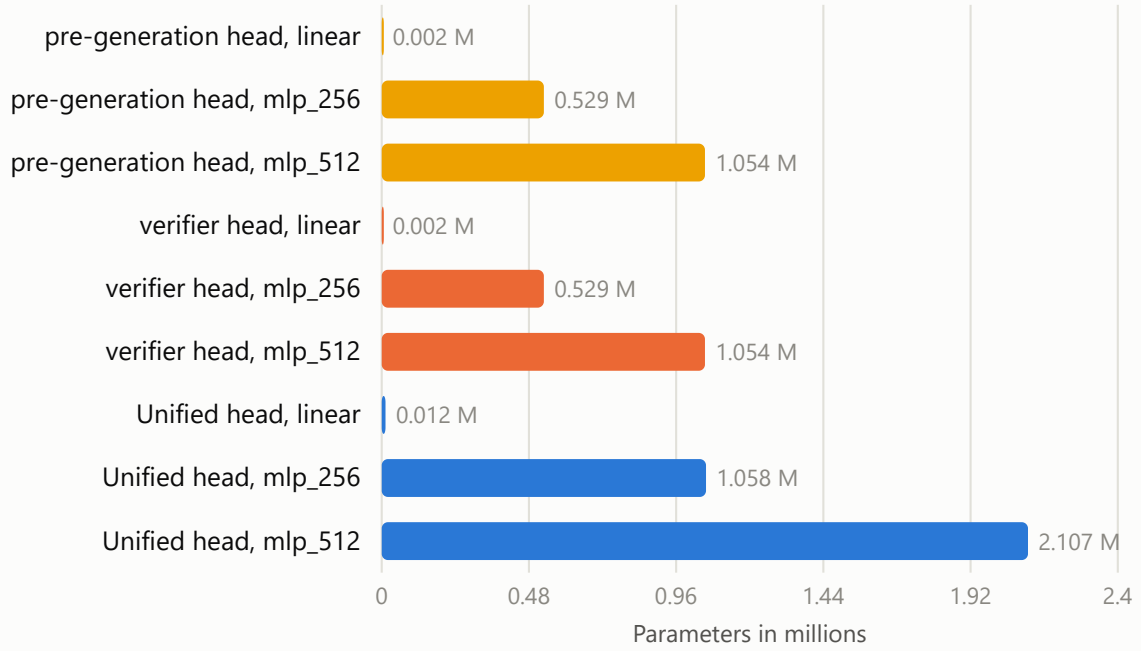
The size of the heads matters for two reasons: it bounds the cost of using them (they add almost nothing to inference) and it shows that the signal is cheap to read. The counts below are exact, computed from the layer sizes for the 2,048-dimensional hidden state of Qwen3-VL-2B. For scale, the language model has roughly 2 billion parameters, so the largest head here is about 0.1% of it.

5.1 Architectures

- **Linear probe** (pre-generation or verifier): $s = w^\top h + b$, with $d + 1$ parameters.
- **MLP-k** (pre-generation or verifier): LayerNorm $(2d)$, Linear $d \to k$, GELU, Dropout 0.1, Linear $k \to 1$. The parameter count is $2d + dk + k + k + 1$.
- **Unified head**: a separate LayerNorm on each half ($4d$ parameters), then the concatenation $[\,\mathrm{LN}(h_q) \,\|\, \mathrm{LN}(h_a) \,\|\, f\,]$ of size $2d + 1$ (the last entry is a flag), then a linear layer or the same MLP body. The count is $4d + (2d + 1)k + k + k + 1$ for an MLP.

Head	Architecture	Input size	Parameters	Megabytes (fp32)	Share of the 2B model
Separate pre-generation head	linear	2,048	2,049	0.01	0.000%
Separate pre-generation head	mlp_256	2,048	528,897	2.12	0.026%
Separate pre-generation head	mlp_512	2,048	1,053,697	4.21	0.053%
Separate verifier head	linear	2,048	2,049	0.01	0.000%
Separate verifier head	mlp_256	2,048	528,897	2.12	0.026%
Separate verifier head	mlp_512	2,048	1,053,697	4.21	0.053%
Unified head	linear	4,096	12,290	0.05	0.001%
Unified head	mlp_256	4,096	1,057,537	4.23	0.053%
Unified head	mlp_512	4,096	2,106,881	8.43	0.105%

Parameters of every head architecture, in millions



A separate MLP-512 has about 1.05 million parameters. The unified MLP-512 has 2.11 million. The linear probes are 2,049 to 12,290 parameters.

The unified head is about twice the size of a separate MLP because it reads two states at once. Its 2.11 million parameters are the same as a separate pre-generation head and a verifier of the same width together (2.11 million), so serving one unified head costs about the same as serving the two separate ones.

5.2 What was chosen in each experiment

No single architecture wins everywhere. Linear probes win the pre-generation and unified heads on ChartQA and the verifier on DocVQA, and the other winners use a hidden layer. Differences between architectures are usually under 0.01 AUROC, so size is not what decides the result. Parameter counts are in brackets.

Dataset	Pre-generation head	Verifier head	Unified head
ChartQA	linear (2,049)	mlp_512 (1,053,697)	linear (12,290)
TextVQA	mlp_256 (528,897)	mlp_256 (528,897)	mlp_256 (1,057,537)
DocVQA	mlp_256 (528,897)	linear (2,049)	mlp_512 (2,106,881)
ScienceQA	mlp_512 (1,053,697)	mlp_256 (528,897)	mlp_512 (2,106,881)

Pooled mixture winners (Section 8). The new loss weights are the grounding weight λ and the pair weight w_{pair}.

Head	Loss	Architecture	Learning rate	Parameters	λ	w_{pair}
Separate pre-generation head	old loss	mlp_256	0.0003	528,897	0	0
Separate pre-generation head	new loss	mlp_512	0.001	1,053,697	0.1	0
Separate verifier head	old loss	mlp_512	0.0003	1,053,697	0	0
Separate verifier head	new loss	mlp_512	0.0003	1,053,697	0.1	0
Unified head	old loss	mlp_512	0.0003	2,106,881	0	0
Unified head	new loss	mlp_512	0.0003	2,106,881	0.1	0

6. Stage 3: three more datasets

Why. A result on one dataset can be an accident of that dataset. ChartQA questions are templated, the answers are numbers, and the model's mistakes may have a simple structure. If the heads only worked there, the project would have a ChartQA tool and not a general gate. Three datasets were added, chosen because the model has to do something different in each: read text in a photograph, read a scanned page, and reason over a diagram with a closed answer set. Each has its own correctness rule, grouped splits, and an offline-tested scoring module. Full details are in report 02.

ChartQA

What it asks. Questions about bar, line, and pie charts, written by people and generated by templates. Answers are numbers or short text read or computed from the chart.

Source. HuggingFaceM4/ChartQA. **Correct means.** Relaxed accuracy: a numeric answer within 5% of the reference counts as correct, text answers need a normalized exact match.

Why it is here. The dataset of the original study, kept as the common reference point.

Splits. Official train, validation, and test splits.

TextVQA

What it asks. Questions that can only be answered by reading text inside a photograph: the brand on a watch, the number on a jersey, what a shirt says. Each question has 10 human answers, which disagree often enough that correctness is a matter of degree.

Source. lmms-lab/textvqa (TextVQA, photographs from Open Images). **Correct means.** Official VQA soft accuracy over the 10 answers. An answer is labeled correct when the score is 0.5 or more, which needs at least two annotators to agree with it.

Why it is here. Tests a different visual skill (reading small, partial, or stylized text in clutter) and free-form answers with natural annotator disagreement.

Splits. 8,000 training questions sampled from the official train split (34,602). Validation (1,500) and test (2,500) are carved out of the labeled official validation split by image, because official test answers are hidden.

DocVQA

What it asks. Questions about scanned pages from industry documents: letters, forms, reports, tables, handwritten notes. Answers are spans copied from the page. Each question carries a type such as table or list, layout, form, free text, handwritten, or figure or diagram.

Source. HuggingFaceM4/DocumentVQA (DocVQA, UCSF Industry Documents Library). **Correct means.** ANLS similarity (one minus normalized edit distance to the closest reference). An answer is labeled correct at 0.9 or more, stricter than the 0.5 cutoff used in the official leaderboard.

Why it is here. Dense text at high resolution, long documents, and answers that must match a span exactly, which is very unlike a chart or a photograph.

Splits. 8,000 training questions sampled from the official train split (39,463). Validation (1,500) and test (2,500) are carved out of the labeled official validation split by document, because official test answers are hidden. Page images are capped at 1,536 pixels on the long side.

ScienceQA

What it asks. Grade-school science questions with a picture or diagram, such as maps, genetics crosses, animal classification, experiments, and reading passages. Every question is multiple choice and the model answers with an option letter.

Source. derek-thomas/ScienceQA (image questions only). **Correct means.** Exact match on the option letter, parsed from the first letter of the reply.

Why it is here. A closed answer set, so the answer itself carries almost no information. It tests whether a correctness head still helps when there is little to read in the answer.

Splits. Official train, validation, and test splits, restricted to questions that have an image.

6.1 Protocol

The same recipe was run per dataset: build the grouped splits, generate the baseline answers, cache the three hidden states, run the separate-head sweep (12 configurations per head, three seeds each) and the unified-head sweep (36 configurations, three seeds, early stopping, maximum 50 epochs, patience 5), and analyze the winners. ChartQA was rerun end to end under this protocol as a pipeline check; it reproduced the original numbers (answer-token probability AUROC 0.751 against 0.750, linear probe 0.895 against 0.895).

6.2 Results

All numbers are on each dataset's test split. The unified head rows are means over three seeds.

Correctness AUROC

Scorer	ChartQA	TextVQA	DocVQA	ScienceQA
Answer-token probability	0.751	0.726	0.685	0.654
Separate pre-gen head	0.895	0.816	0.776	0.918
Separate verifier head	0.892	0.868	0.788	0.879
Unified, pre-gen mode	0.894	0.812	0.769	0.923
Unified, joint mode	0.898	0.867	0.811	0.925

Error AUPRC

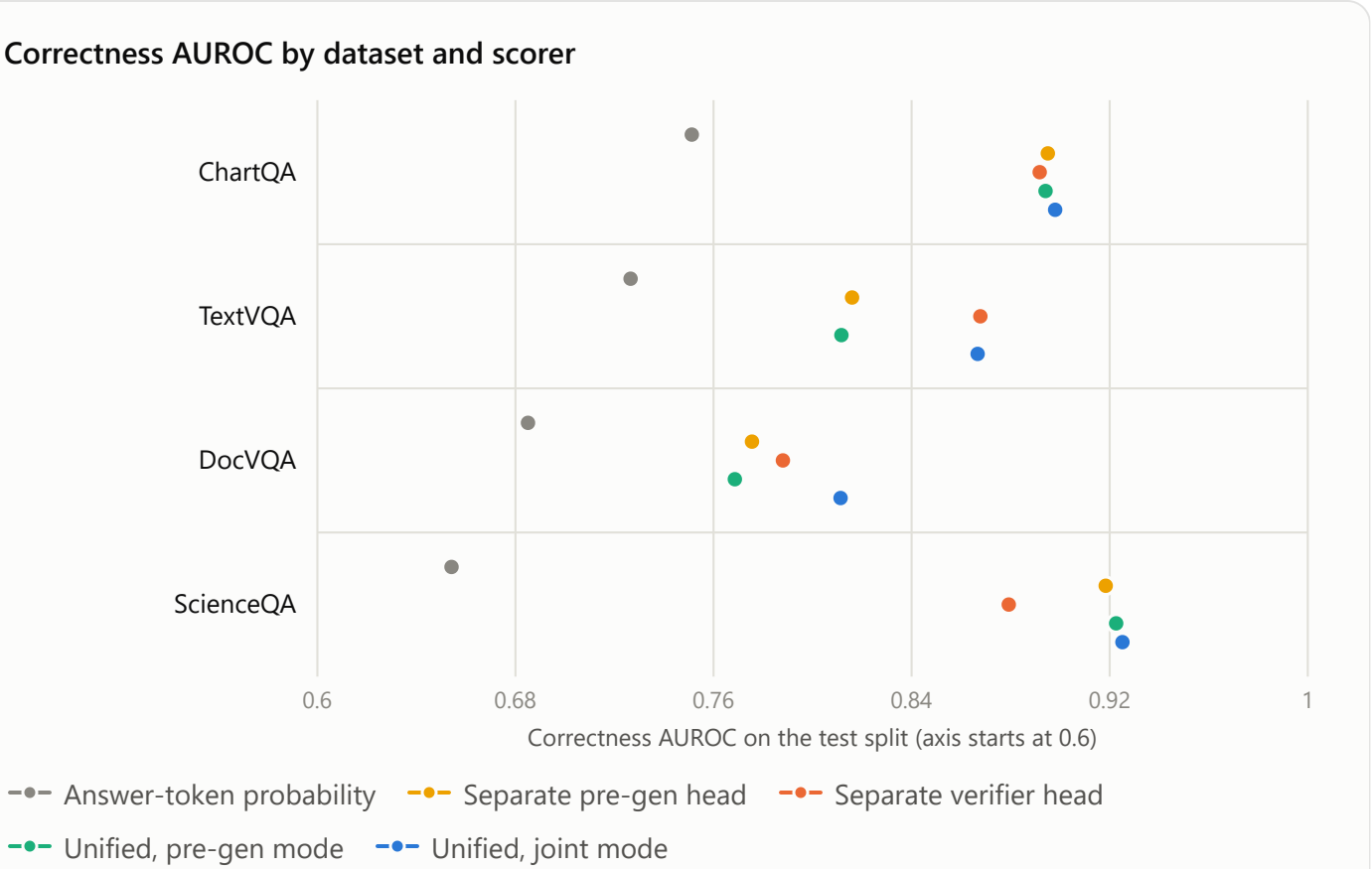
Scorer	ChartQA	TextVQA	DocVQA	ScienceQA
Answer-token probability	0.581	0.449	0.333	0.323
Separate pre-gen head	0.732	0.527	0.399	0.560
Separate verifier head	0.732	0.644	0.443	0.503
Unified, pre-gen mode	0.741	0.514	0.384	0.573
Unified, joint mode	0.751	0.635	0.462	0.577

Brier score (lower is better)

Scorer	ChartQA	TextVQA	DocVQA	ScienceQA
Answer-token probability	0.180	0.158	0.114	0.116
Separate pre-gen head	0.108	0.122	0.096	0.078
Separate verifier head	0.110	0.103	0.094	0.084
Unified, pre-gen mode	0.108	0.123	0.097	0.073
Unified, joint mode	0.107	0.107	0.090	0.072

ECE (lower is better)

Scorer	ChartQA	TextVQA	DocVQA	ScienceQA
Answer-token probability	0.175	0.152	0.106	0.108
Separate pre-gen head	0.026	0.042	0.015	0.036
Separate verifier head	0.032	0.016	0.013	0.023
Unified, pre-gen mode	0.031	0.034	0.020	0.019
Unified, joint mode	0.038	0.037	0.014	0.021



READING THE NUMBERS

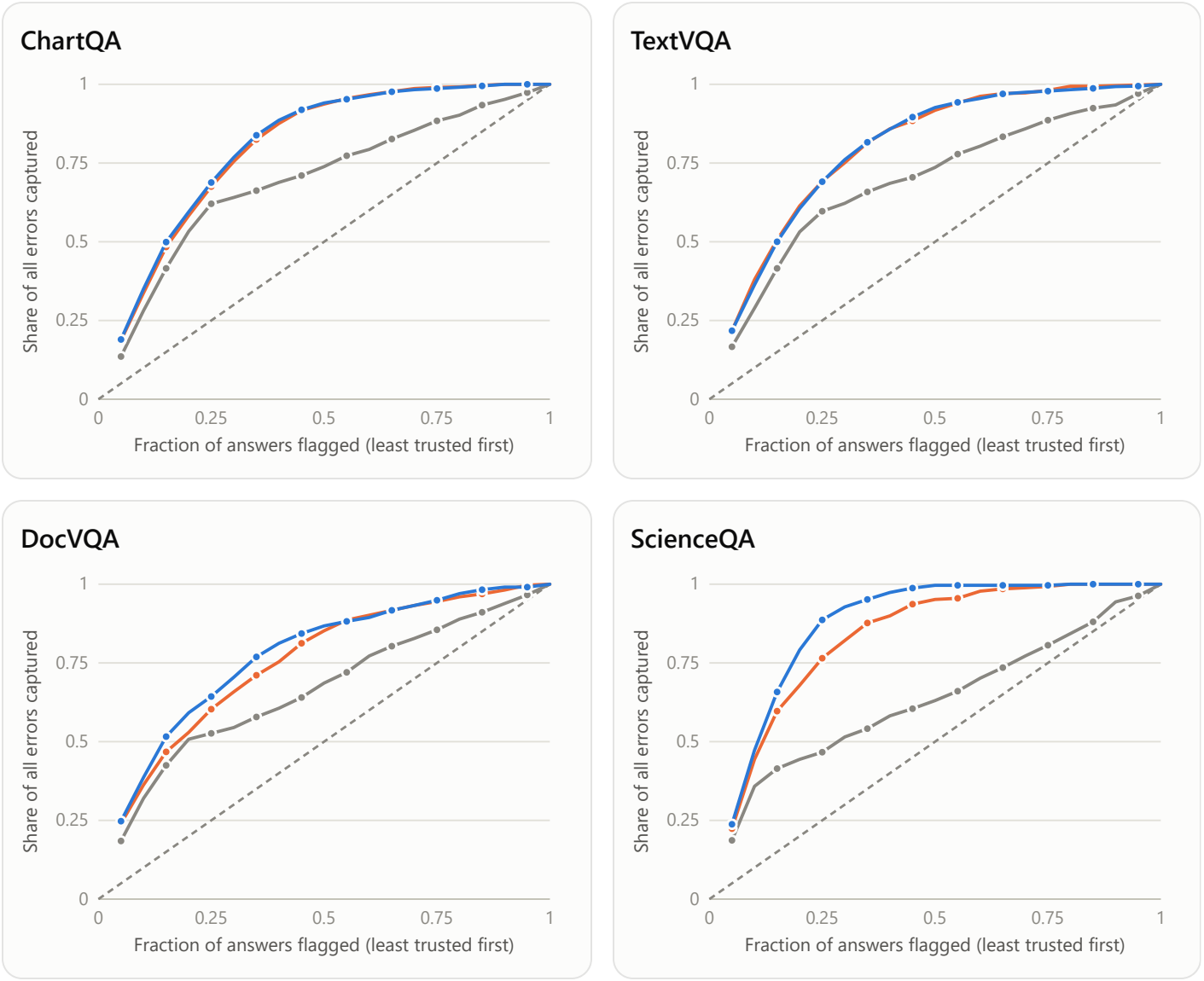
The learned heads beat the answer-token probability on every dataset and every metric. The gain in AUROC of the unified joint head runs from 0.13 (DocVQA) to 0.27 (ScienceQA). Calibration improves most: ECE falls from the range 0.11 to 0.18 for the probability to between 0.01 and 0.04 for the heads, so the head's number can be read as a probability. The separate verifier and the unified joint head are close everywhere. The pre-generation scores sit lower than the answer-aware ones on TextVQA and DocVQA, where the answer text carries the signal, and close to them on ScienceQA, where the question alone tells the head which topics are hard.

6.3 Using the score as a gate

If a correction step can only act on a fraction of answers, what matters is how many errors the flagged fraction contains. Flagging the 20% of answers with the lowest score:

Dataset	Errors found, answer-token probability	Errors found, unified head	Accuracy of top half, answer-token probability	Accuracy of top half, unified head	Base accuracy
ChartQA	53%	59%	87.5%	97.1%	75.8%
TextVQA	53%	61%	88.9%	97.2%	81.0%
DocVQA	51%	59%	91.5%	96.6%	87.0%

Dataset	Errors found, answer-token probability	Errors found, unified head	Accuracy of top half, answer-token probability	Accuracy of top half, unified head	Base accuracy
ScienceQA	44%	79%	91.2%	99.9%	86.7%



The dashed diagonal is random flagging. The learned head's curve sits well above the answer-token probability on every dataset.

6.4 Does the head have to be a pair?

Paired bootstrap on the test split (1,000 resamples) gives the difference between the unified head and the separate head of the same kind, as AUROC with a 95% interval. An interval that includes zero means the two are not distinguishable.

Dataset	Unified joint minus separate verifier	Unified pre-generation minus separate pre-generation
ChartQA	+0.0068 (-0.0009 to +0.0147)	-0.0003 (-0.0038 to +0.0031)
TextVQA	-0.0006 (-0.0111 to +0.0112)	-0.0028 (-0.0067 to +0.0011)

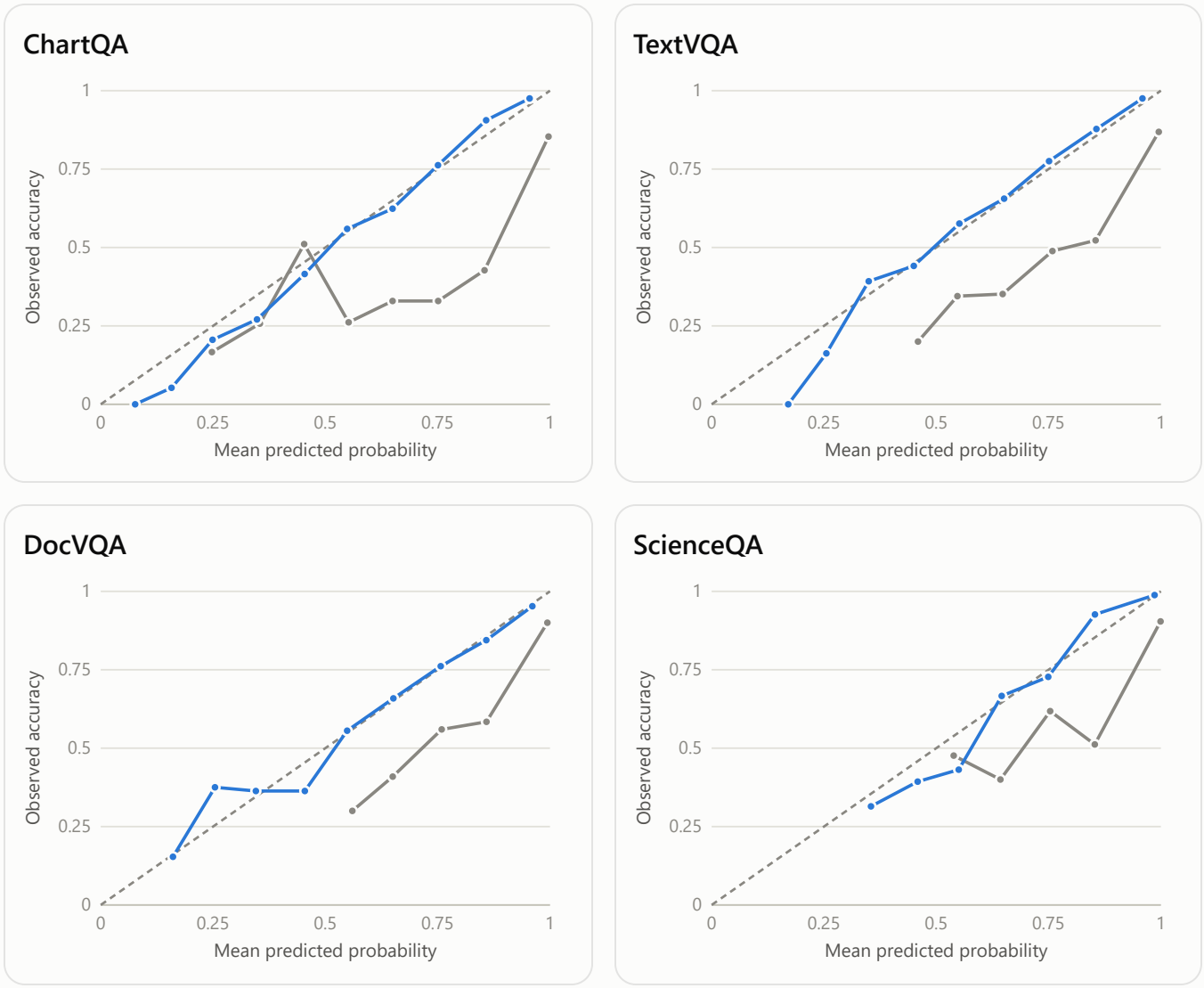
Dataset	Unified joint minus separate verifier	Unified pre-generation minus separate pre-generation
DocVQA	+0.0238 (+0.0027 to +0.0450)	-0.0057 (-0.0110 to -0.0009)
ScienceQA	+0.0471 (+0.0339 to +0.0614)	+0.0049 (+0.0010 to +0.0087)

In joint mode the unified head is never worse than the separate verifier, and its interval excludes zero in its favor on DocVQA and ScienceQA (on ChartQA and TextVQA the interval includes zero). In pre-generation mode it is within 0.006 AUROC of the separate head: slightly better on ScienceQA, slightly lower on DocVQA, and indistinguishable on the other two. This is why the unified head is kept: it matches two separate heads on the whole and it is one network.

6.5 Calibration

A score is useful as a probability only if it matches how often answers are right. The charts plot, for bins of predicted probability, the observed accuracy of the answers in the bin (bins with fewer than 10 answers are dropped). The dashed diagonal is perfect calibration. The answer-token probability is over-confident: in most bins it sits well below the diagonal, and answers it scores between 0.6 and 0.9 are right only about 30% to 60% of the time. The unified head follows the diagonal closely, which is why its ECE is low.

Reliability of the answer-token probability and the unified head



Test split.

6.6 The gate in more detail

Share of all errors found when flagging a given fraction of the least trusted answers (a random flag finds the same share as the flag rate):

Dataset	Flag rate	Answer-token probability	Separate verifier	Unified joint
ChartQA	5%	14%	19%	19%
ChartQA	10%	28%	34%	35%
ChartQA	20%	53%	58%	59%
ChartQA	30%	64%	75%	77%
ChartQA	50%	74%	94%	94%

Dataset	Flag rate	Answer-token probability	Separate verifier	Unified joint
TextVQA	5%	17%	22%	22%
TextVQA	10%	29%	38%	36%
TextVQA	20%	53%	61%	61%
TextVQA	30%	62%	75%	76%
TextVQA	50%	74%	92%	93%
DocVQA	5%	18%	24%	25%
DocVQA	10%	32%	36%	39%
DocVQA	20%	51%	53%	59%
DocVQA	30%	54%	66%	70%
DocVQA	50%	69%	85%	87%
ScienceQA	5%	19%	22%	24%
ScienceQA	10%	36%	44%	47%
ScienceQA	20%	44%	68%	79%
ScienceQA	30%	51%	82%	93%
ScienceQA	50%	63%	95%	100%

Accuracy of the answers kept when only the most trusted fraction is kept (answer-token probability / unified head):

Dataset	Base accuracy	Keep 90%	Keep 80%	Keep 50%	Keep 20%
ChartQA	75.8%	80.7% / 82.6%	85.9% / 87.8%	87.5% / 97.1%	88.4% / 98.9%
TextVQA	81.0%	85.1% / 86.6%	88.8% / 90.6%	88.9% / 97.2%	88.6% / 98.4%
DocVQA	87.0%	90.2% / 91.1%	91.9% / 93.4%	91.5% / 96.6%	92.8% / 98.1%
ScienceQA	86.7%	90.5% / 92.2%	90.5% / 96.5%	91.2% / 99.9%	90.1% / 100.0%

6.7 Question types

Where the dataset labels question types, accuracy and the unified head's mean predicted probability by type. A close match means the head knows which types are hard. The head is well matched for most types and over-confident on handwritten pages (74% accurate against 83% predicted).

Dataset	Question type	Questions	Accuracy	Mean predicted probability
DocVQA	table/list	753	87.4%	87.4%
DocVQA	layout	698	89.8%	90.1%
DocVQA	form	374	94.4%	92.5%
DocVQA	free_text	352	80.4%	82.3%
DocVQA	handwritten	149	73.8%	82.6%
DocVQA	figure/diagram	111	81.1%	84.5%

Dataset	Question type	Questions	Accuracy	Mean predicted probability
ScienceQA	natural science	1,209	81.5%	82.6%
ScienceQA	social science	764	94.6%	94.4%
ScienceQA	language science	44	93.2%	91.2%

6.8 Examples

Two examples per group per dataset from the test split. **Caught error**: a wrong answer the unified head scored low. **Missed error**: a wrong answer it scored high. **False alarm**: a correct answer it scored low. **Confident and correct**: a correct answer it scored high. The unified score is the head's probability that the answer is right.

Dataset	Group	Question	Reference	Model answer	Unified score	Answer-token prob.
ChartQA	Caught error	Use blue bar data and grey bar data, find top highest number from each group and find the average of all th...	70	To find the top highest number from each group ...	0.02	0.86
ChartQA	Caught error	Add the three largest government procurement in GDP and multiply it by the smallest government procurement ...	389.82	(17.28 + 15.05 + 12.17) * 8.76 = 44.5 * 8.76 = ...	0.03	0.92
ChartQA	Missed error	What was Vodafone's revenue in Italy in 2021?	5857	5014	0.97	1.00
ChartQA	Missed error	What was the total amount of U.S. exports to the EU in 2020?	336.73	237.9	0.97	1.00
ChartQA	False alarm	What is the difference between maximum gross profit margin of ASOS and minimum gross profit margin of Zalando?	5	5	0.11	0.91

Dataset	Group	Question	Reference	Model answer	Unified score	Answer-token prob.
ChartQA	False alarm	What is the difference in the value of both the bars?	7.1	7.1	0.12	0.80
ChartQA	Confident and correct	What was the female population of Madagascar in 2019?	13.52	13.52	0.99	1.00
ChartQA	Confident and correct	What was the crude birth rate in Sudan in 2018?	32.18	32.18	0.99	1.00
TextVQA	Caught error	what does it say on the ladies shirts?	volcvo	volcavolcavolcavolcavolcavolcavolcavolcavo...	0.11	0.99
TextVQA	Caught error	what buttons are on the remote?	vol, channel, menu, record	vol, menu, enter, ch, delete, mute, q.menu, lis...	0.12	0.88
TextVQA	Missed error	is this live or not?	yes	live	0.99	1.00
TextVQA	Missed error	what is the band name on this t-shirt?	ab cd	acdc	0.99	1.00
TextVQA	False alarm	what are numbers on this remote?	1234567890	1 2 3 4 5 6 7 8 9 0	0.21	1.00
TextVQA	False alarm	what time is shown on the watch?	10:10	10:10	0.22	0.86
TextVQA	Confident and correct	what is the brand of this watch?	seiko	seiko	1.00	1.00
TextVQA	Confident and correct	is there a player with a 6 number?	yes	yes	1.00	1.00
DocVQA	Caught error	Which compounds are used as substitutes for sodium compounds?	potassium compounds	potassium compounds, sodium benzoate, sodium ch...	0.08	0.96
DocVQA	Caught error	What are the dates written near L.A.?	4-5-6	30.1-2 Aug	0.11	0.77
DocVQA	Missed error	What is the title of the given bar graph?	average age of product line	Average Age of Product Line 2004 - 2014	0.99	1.00

Dataset	Group	Question	Reference	Model answer	Unified score	Answer-token prob.
DocVQA	Missed error	With how many parts were the'tests accomplished'?	24	24 parts	0.99	0.88
DocVQA	False alarm	What is the handwritten instruction regarding what to do with the flyer?	DON'T PUT THIS IN THE WASTE BASKET SE...	Don't put the in the waste basket, send it to t...	0.16	0.81
DocVQA	False alarm	If you start from top left, who is the first person with more than one abstract number associated with him/...	Abdel-Hak, T.M.	Abdul-Hak, T.M.	0.17	0.87
DocVQA	Confident and correct	What is written in the Division field ?	nutrition	Nutrition	1.00	1.00
DocVQA	Confident and correct	What is the 4th concern?	quantification of the suggestive canc...	Quantification of the Suggestive cancer endpoint	1.00	1.00
ScienceQA	Caught error	Context: In a group of pea plants, some individuals have green pods and others have yellow pods. In this gr...	C	B. 4:0	0.26	1.00
ScienceQA	Caught error	Context: In a group of rabbits, some individuals have black fur and others have brown fur. In this group, t...	D	E	0.26	1.00
ScienceQA	Missed error	Context: Mammals have hair or fur and feed their young milk. Mammals are warm-blooded. Warm-blooded animals...	B	C	1.00	1.00

Dataset	Group	Question	Reference	Model answer	Unified score	Answer-token prob.
ScienceQA	Missed error	Question: What is the capital of North Dakota? Options: A. Pierre B. Juneau C. Bismarck D. Jefferson City A...	C	A	0.98	1.00
ScienceQA	False alarm	Context: In a group of horses, some individuals have a black coat and others have a reddish-brown coat. In ...	E	E	0.21	0.51
ScienceQA	False alarm	Context: This passage describes the wing type trait in fruit flies: Some scientists breed fruit flies to le...	E	E	0.26	1.00
ScienceQA	Confident and correct	Question: Which continent is highlighted? Options: A. Africa B. Europe C. South America D. North America An...	A	A	1.00	1.00
ScienceQA	Confident and correct	Question: Which continent is highlighted? Options: A. Africa B. North America C. South America D. Europe An...	A	A	1.00	1.00

The caught errors show what the head is picking up: overlong or looping generations (a repeated string on a TextVQA question, a long list of buttons), chart arithmetic written out step by step, and answers that name the wrong option. Two kinds of missed error appear. Some are confident misreads, such as a wrong value read from a chart, which a head that only sees hidden states has no way to check. Others are answers that are right in spirit and marked wrong by the strict label rule ("live" for a question whose reference is "yes", "24 parts" against "24"), so part of the missed-error count reflects label noise and not head failure. The false alarms show the reverse: correct answers in an unusual format ("1 2 3 4 5 6 7 8 9 0" for "1234567890") that the head distrusts.

6.9 A length shortcut check

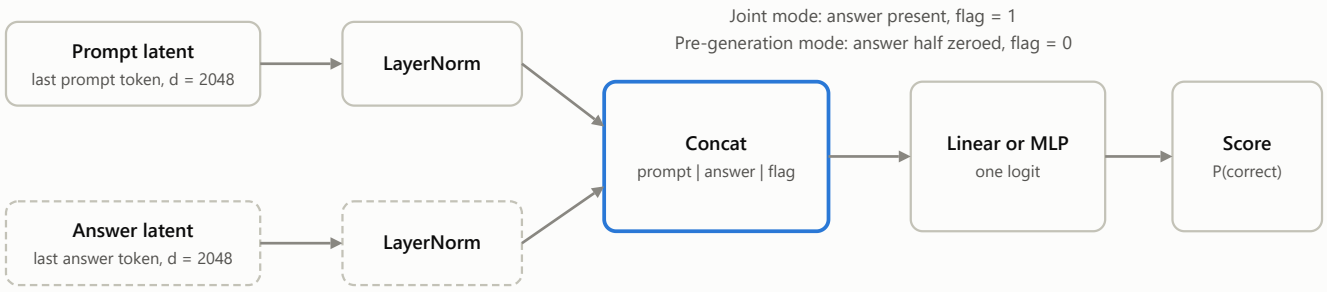
If wrong answers were simply longer, a head could score well by counting characters. An answer-length score alone (length in characters, shorter answers scored as more likely correct) reaches only the AUROCs below, where 0.5 is chance and the heads reach 0.78 to 0.92, so the heads are not using length as a shortcut. TextVQA is the one dataset where length carries some signal (wrong answers average 13.7 characters against 7.8), and the heads still beat it by more than 0.25 AUROC.

Dataset	Answer-length AUROC	Answer-length error AUPRC	Mean length, correct	Mean length, wrong
ChartQA	0.420	0.240	5.1	6.6
TextVQA	0.612	0.326	7.8	13.7
DocVQA	0.507	0.182	13.9	19.9
ScienceQA	0.526	0.178	1.4	2.8

7. Stage 4: the unified head

The idea. Two separate heads need two networks, two trainings and two decisions about which to trust. A single head that reads both states removes that. It also gives a natural way to combine them: the question state h_q gives the difficulty prior, and the answer state h_a gives the evidence about this particular answer.

The unified head



Two normalized halves, one flag, and one output. With the answer half zeroed it is a pre-generation scorer.

7.1 Design

Each half is normalized separately, so neither dominates by scale. The input is the concatenation and a flag f:

$$x = \left[\, \mathrm{LN}_q(h_q) \,\|\, \mathrm{LN}_a(h_a) \,\|\, f \,\right], \qquad s_{\text{joint}} = g(x),\ f = 1$$

$$s_{\text{pre}} = g\big(\left[\, \mathrm{LN}_q(h_q) \,\|\, \mathbf{0} \,\|\, 0 \,\right]\big)$$

With the answer half zeroed and the flag at 0, the same network scores a question before any answer exists (**pre-generation mode**). With both halves present it scores a candidate answer (**joint mode**). This is trained jointly with a loss that keeps both modes useful:

$$L = L_{\mathrm{BCE}}(s_{\mathrm{joint}}, y) + w_{\mathrm{pre}}\, L_{\mathrm{BCE}}(s_{\mathrm{pre}}, y) + w_{\mathrm{rank}} L_{\mathrm{rank}} + w_{\mathrm{pair}} L_{\mathrm{pair}}$$

The reason for the pre-generation term is that a head trained only on the joint input would learn to rely on the answer half and could be useless when the answer half is empty. The flag and the zeroing teach it a separate, honest mode for the question-only case.

7.2 Results

Test AUROC per dataset, separate heads against the unified head in the same mode.

Dataset	Separate pre-generation	Unified, pre-generation mode	Separate verifier	Unified, joint mode
ChartQA	0.895	0.894	0.892	0.898
TextVQA	0.816	0.812	0.868	0.867
DocVQA	0.776	0.769	0.788	0.811
ScienceQA	0.918	0.923	0.879	0.925

> **FINDINGS**
> The unified head's pre-generation mode matches the separate pre-generation head to within 0.007 AUROC on every dataset, and its joint mode matches or beats the separate verifier on all four datasets (the largest gains are on DocVQA and ScienceQA). The pre-generation loss term costs almost nothing: it lifts the question-only mode on every dataset and moves the joint mode by at most 0.006 AUROC on validation. The pairwise reference-answer loss lowered validation AUROC on every dataset and is switched off in all winning configurations. One network therefore does the job of two without a measurable loss.

---

8. Stage 5: one head for all four datasets

Why. The per-dataset heads show that the signal exists in each dataset. A deployed gate does not know which dataset a question comes from. The pooled experiment asks whether one head, trained on a mix, scores all of them well, and whether it pays a price against heads trained on a single dataset. Details are in report 05.

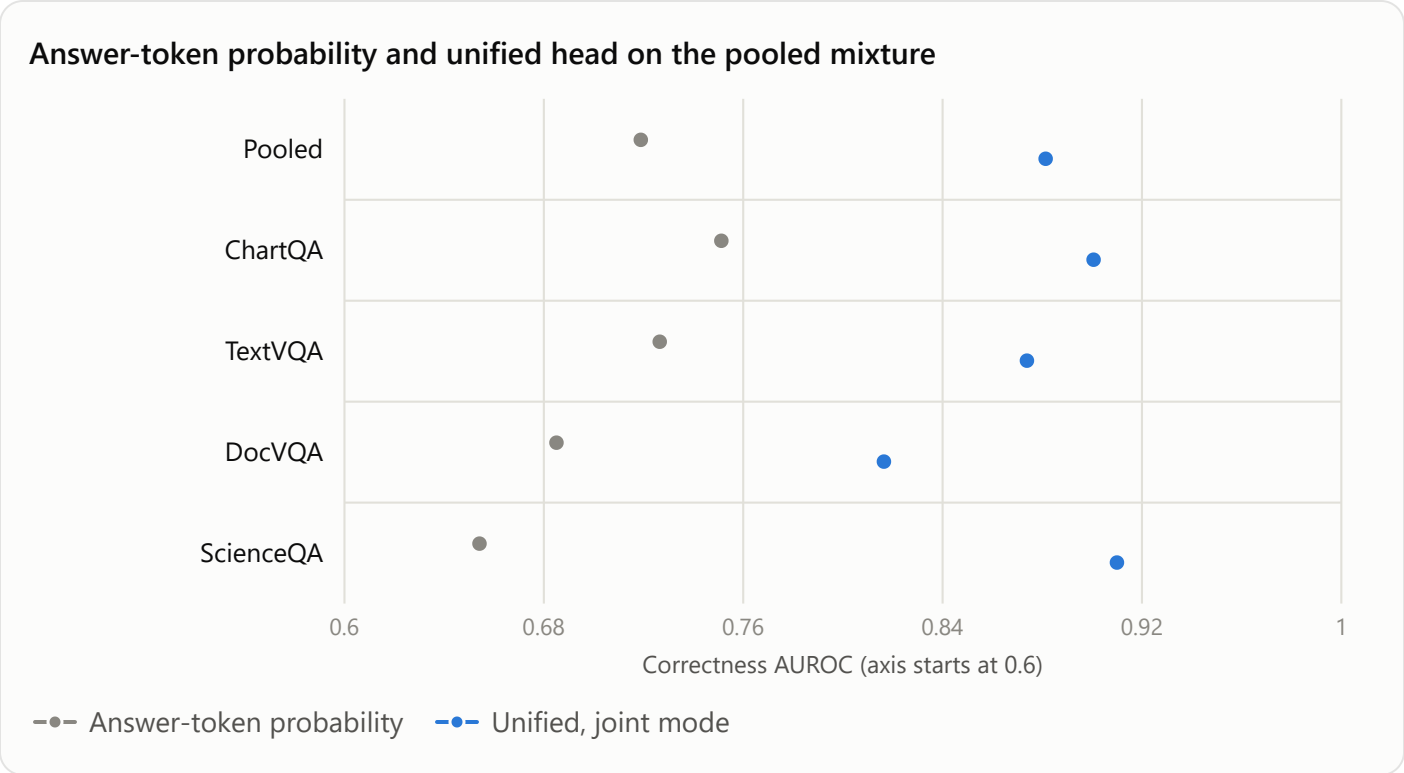
Setup. The mixture is the union of the four datasets' existing splits: 50,517 training rows, 6,420 validation rows, 9,517 test rows, with an overall base accuracy of 82.4% on the test split. Nothing was resampled, so every test question is one the per-dataset study also scored. One calibration temperature is fitted on the pooled validation split. The sweep has 36 head configurations for the pre-generation head and for the verifier and 72 for the unified head, each on three seeds, covering the old loss (no grounding term) and the new loss.

8.1 Experiment 1: the answer-token probability and linear probes

No training is needed for the probability. The linear probes are the best linear heads of the sweep, trained on the pool.

8.2 Experiments 2 and 3: separate and unified heads, old loss

Correctness AUROC	Pooled	ChartQA	TextVQA	DocVQA	ScienceQA
Answer-token probability	0.719	0.751	0.726	0.685	0.654
Linear probe, pre-generation	0.857	0.893	0.821	0.781	0.897
Linear probe, verifier	0.864	0.885	0.859	0.798	0.885
Linear probe, unified joint	0.878	0.896	0.871	0.814	0.904
Separate pre-generation head	0.864	0.894	0.831	0.793	0.903
Separate verifier head	0.873	0.895	0.869	0.807	0.890
Unified, pre-generation mode	0.861	0.893	0.823	0.788	0.906
Unified, joint mode	0.881	0.901	0.874	0.816	0.910



8.3 Generalist against specialist

Is a head trained on a mixture worse on a dataset than a head trained on that dataset alone? The table compares the unified joint head. A positive difference means the pooled head is better.

Dataset	Trained on that dataset alone	Trained on the pool	Difference
ChartQA	0.898	0.901	+0.003
TextVQA	0.867	0.874	+0.007
DocVQA	0.811	0.816	+0.005
ScienceQA	0.925	0.910	-0.015

FINDINGS

One unified head trained on the pool reaches pooled AUROC 0.881 against 0.719 for the answer-token probability. On each dataset it is within 0.015 AUROC of the head trained on that dataset alone, and above it on three of the four, so the pool does not cost much accuracy and on three datasets helps slightly, possibly because the other datasets add training data for the same signal. A linear probe already reaches 0.86 to 0.88 pooled, so most of the signal is linearly readable; the hidden layer and the answer half add the remaining few points.

8.4 The gate on the pooled mixture

Flagging the 20% of pooled answers with the lowest score, and keeping the most trusted 50%:

Scorer	Errors found in the lowest 20%	Accuracy of the most trusted 50%
Answer-token probability	55%	89.8%
Linear probe (unified joint)	65%	97.3%
Unified joint, old loss	65%	97.7%
Unified joint, new loss	65%	97.7%
Separate verifier, old loss	64%	97.2%
Separate pre-generation, old loss	62%	97.0%

8.5 What the sweep says about the hyperparameters

Every sweep fits many configurations, so it is also a controlled experiment on the design choices. Each row averages the validation metric over all configurations that share one value of one factor (with the other factors varying), so it shows the marginal effect of that choice. These are validation numbers from the swap sweep; the noise sweep is in the Excel workbook.

Separate pre-generation head

Factor	Value	Configurations	Mean validation AUROC	Best validation AUROC	Mean validation error AUPRC
architecture	linear	12	0.8445	0.8469	0.5940
architecture	mlp_256	12	0.8523	0.8537	0.6055
architecture	mlp_512	12	0.8523	0.8546	0.6056
learning rate	0.0003	18	0.8497	0.8535	0.6022

Factor	Value	Configurations	Mean validation AUROC	Best validation AUROC	Mean validation error AUPRC
learning rate	0.001	18	0.8497	0.8546	0.6012
rank weight	0.0	18	0.8492	0.8541	0.6011
rank weight	0.25	18	0.8502	0.8546	0.6023
ground weight	0.0	12	0.8507	0.8537	0.6060
ground weight	0.1	12	0.8510	0.8546	0.6052
ground weight	0.5	12	0.8474	0.8520	0.5939

Separate verifier head

Factor	Value	Configurations	Mean validation AUROC	Best validation AUROC	Mean validation error AUPRC
architecture	linear	12	0.8279	0.8493	0.5578
architecture	mlp_256	12	0.8393	0.8558	0.5776
architecture	mlp_512	12	0.8404	0.8573	0.5803
learning rate	0.0003	18	0.8345	0.8573	0.5690
learning rate	0.001	18	0.8372	0.8572	0.5748
pair weight	0.0	18	0.8492	0.8573	0.5965
pair weight	0.25	18	0.8225	0.8334	0.5473
ground weight	0.0	12	0.8388	0.8573	0.5775
ground weight	0.1	12	0.8369	0.8546	0.5745
ground weight	0.5	12	0.8318	0.8502	0.5638

Unified head

Factor	Value	Configurations	Mean validation AUROC	Best validation AUROC	Mean validation error AUPRC
architecture	linear	24	0.8557	0.8663	0.6085
architecture	mlp_256	24	0.8627	0.8711	0.6219
architecture	mlp_512	24	0.8630	0.8711	0.6224
learning rate	0.0003	36	0.8604	0.8711	0.6179
learning rate	0.001	36	0.8605	0.8704	0.6174
pre weight	0.0	36	0.8584	0.8704	0.6136
pre weight	1.0	36	0.8625	0.8711	0.6217
pair weight	0.0	36	0.8667	0.8711	0.6286
pair weight	0.25	36	0.8542	0.8607	0.6067
ground weight	0.0	24	0.8604	0.8711	0.6193
ground weight	0.1	24	0.8615	0.8711	0.6200
ground weight	0.5	24	0.8595	0.8688	0.6136

Three things stand out. First, the learning rate barely matters (under 0.003 AUROC). Second, the architecture matters a little: the linear probe averages 0.007 to 0.013 AUROC below the MLPs, and MLP-256 and MLP-512 are indistinguishable, which is why the linear probe is so close to the larger heads in the winners. Third, the grounding weight λ and the pair weight are the factors that change the picture: a pair weight of 0.25 lowers mean validation AUROC by 0.013 (unified) to 0.027 (verifier), while λ costs 0.003 or less at 0.1 and up to 0.007 at 0.5. This is why the winning new-loss configurations use $\lambda = 0.1$ and no pair term.

8.6 How much does a seed matter?

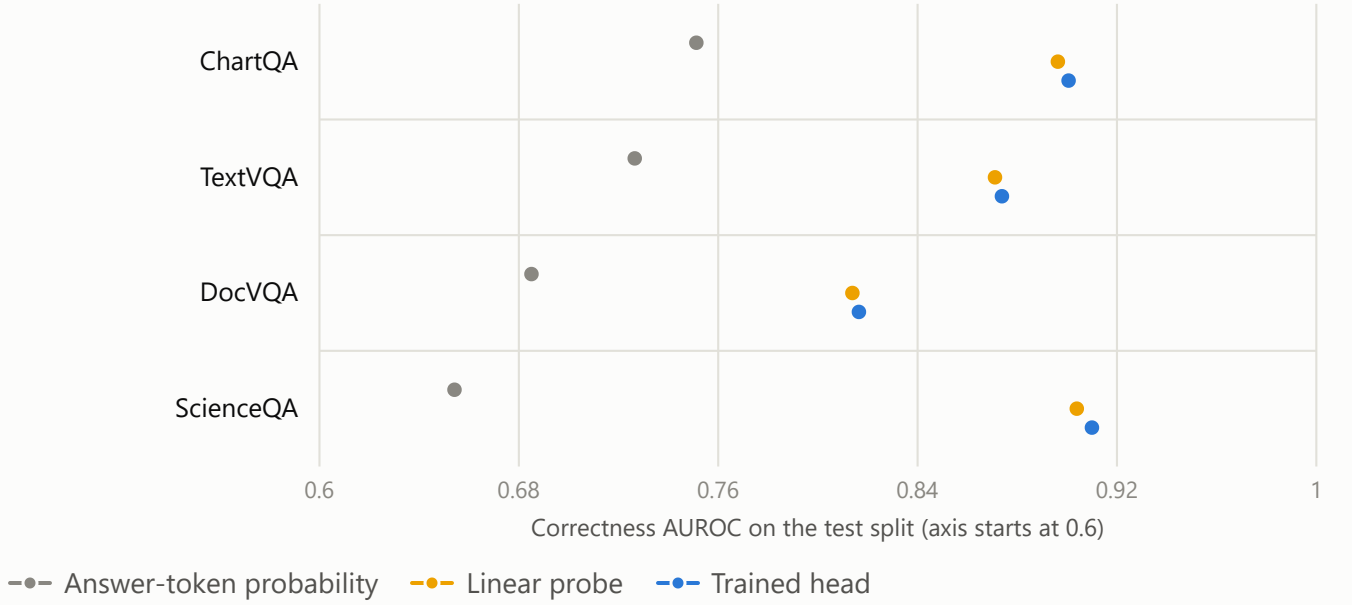
Standard deviation of the pooled test AUROC across the three seeds of each winner. Differences smaller than about twice these values should not be read as real.

Head	Loss	Pooled AUROC	Std over seeds
Separate pre-generation head	old loss	0.8643	0.0004
Separate pre-generation head	new loss	0.8625	0.0018
Separate verifier head	old loss	0.8727	0.0007
Separate verifier head	new loss	0.8714	0.0008
Unified head, pre-generation	old loss	0.8614	0.0023
Unified head, pre-generation	new loss	0.8606	0.0018
Unified head, joint	old loss	0.8813	0.0024
Unified head, joint	new loss	0.8822	0.0005

8.7 Trained heads against linear probes, per dataset

The question here is how much the extra capacity buys. A **linear probe** is the best linear head of a sweep: no hidden layer, 2,049 parameters for a separate head and 12,290 for the unified head. The **trained head** is the best head of the same sweep over all three architectures (linear, MLP-256, MLP-512), and it is a linear probe whenever the validation score prefers one. Both are chosen on validation with the old loss and scored on the test split, mean and standard deviation over three seeds. The comparison is made in the two training settings used in this report, for every head kind and mode. A difference is called real only if it is larger than the combined seed spread of the two. The answer-token probability is shown as a reference.

Unified head, joint mode: answer-token probability, linear probe and trained head (trained on the pool)



Each row is one dataset's test rows.

- **Trained on the pooled mixture.** The trained head is better than the linear probe by more than the seed spread in 12 of 16 comparisons, within the spread (or the same model) in 4, and worse in 0. The mean AUROC difference is +0.0061 and the largest is +0.013. The linear probe delivers 88% to 99% (median 97%) of the head's gain over the answer-token probability.
- **Trained on each dataset alone.** The trained head is better than the linear probe by more than the seed spread in 7 of 16 comparisons, within the spread (or the same model) in 7, and worse in 2. The mean AUROC difference is +0.0027 and the largest is +0.009. The linear probe delivers 91% to 106% (median 98%) of the head's gain over the answer-token probability.

Trained on the pooled mixture

Dataset	Head	Answer-token prob. AUROC	Linear probe AUROC	Trained head AUROC	Difference	Head architecture (parameters)	Linear probe error AUPRC	Trained head error AUPRC	Reading
ChartQA	Pre-generation head	0.751	0.893 ± 0.000	0.894 ± 0.001	+0.002	mlp_256 (528,897)	0.729	0.742	head better
ChartQA	Verifier head	0.751	0.885 ± 0.000	0.895 ± 0.001	+0.010	mlp_512 (1,053,697)	0.703	0.740	head better
ChartQA	Unified, pre-generation mode	0.751	0.892 ± 0.000	0.893 ± 0.003	+0.001	mlp_512 (2,106,881)	0.735	0.734	within seed spread
ChartQA	Unified, joint mode	0.751	0.896 ± 0.000	0.901 ± 0.002	+0.004	mlp_512 (2,106,881)	0.744	0.752	head better

Dataset	Head	Answer-token prob. AUROC	Linear probe AUROC	Trained head AUROC	Difference	Head architecture (parameters)	Linear probe error AUPRC	Trained head error AUPRC	Reading
TextVQA	Pre-generation head	0.726	0.821 ± 0.002	0.831 ± 0.001	+0.010	mlp_256 (528,897)	0.545	0.556	head better
TextVQA	Verifier head	0.726	0.859 ± 0.003	0.869 ± 0.001	+0.010	mlp_512 (1,053,697)	0.636	0.653	head better
TextVQA	Unified, pre-generation mode	0.726	0.821 ± 0.000	0.823 ± 0.008	+0.002	mlp_512 (2,106,881)	0.545	0.537	within seed spread
TextVQA	Unified, joint mode	0.726	0.871 ± 0.000	0.874 ± 0.003	+0.003	mlp_512 (2,106,881)	0.653	0.657	within seed spread
DocVQA	Pre-generation head	0.685	0.781 ± 0.011	0.793 ± 0.004	+0.013	mlp_256 (528,897)	0.402	0.418	head better
DocVQA	Verifier head	0.685	0.798 ± 0.002	0.807 ± 0.001	+0.009	mlp_512 (1,053,697)	0.449	0.459	head better
DocVQA	Unified, pre-generation mode	0.685	0.782 ± 0.002	0.788 ± 0.002	+0.006	mlp_512 (2,106,881)	0.402	0.409	head better
DocVQA	Unified, joint mode	0.685	0.814 ± 0.002	0.816 ± 0.006	+0.003	mlp_512 (2,106,881)	0.466	0.480	within seed spread
ScienceQA	Pre-generation head	0.654	0.897 ± 0.000	0.903 ± 0.002	+0.006	mlp_256 (528,897)	0.532	0.554	head better
ScienceQA	Verifier head	0.654	0.885 ± 0.001	0.890 ± 0.001	+0.005	mlp_512 (1,053,697)	0.514	0.538	head better
ScienceQA	Unified, pre-generation mode	0.654	0.898 ± 0.000	0.906 ± 0.003	+0.008	mlp_512 (2,106,881)	0.535	0.559	head better
ScienceQA	Unified, joint mode	0.654	0.904 ± 0.001	0.910 ± 0.003	+0.006	mlp_512 (2,106,881)	0.546	0.563	head better

Trained on each dataset alone

Dataset	Head	Answer-token prob. AUROC	Linear probe AUROC	Trained head AUROC	Difference	Head architecture (parameters)	Linear probe error AUPRC	Trained head error AUPRC	Reading
ChartQA	Pre-generation head	0.751	0.893 ± 0.002	0.896 ± 0.001	+0.003	mlp_512 (1,053,697)	0.727	0.745	head better

Dataset	Head	Answer-token prob. AUROC	Linear probe AUROC	Trained head AUROC	Difference	Head architecture (parameters)	Linear probe error AUPRC	Trained head error AUPRC	Reading
ChartQA	Verifier head	0.751	0.886 ± 0.000	0.894 ± 0.002	+0.007	mlp_256 (528,897)	0.709	0.736	head better
ChartQA	Unified, pre-generation mode	0.751	0.895 ± 0.000	0.895 ± 0.000	+0.000	linear (12,290)	0.741	0.741	same model (the best head is linear)
ChartQA	Unified, joint mode	0.751	0.898 ± 0.000	0.898 ± 0.000	+0.000	linear (12,290)	0.750	0.750	same model (the best head is linear)
TextVQA	Pre-generation head	0.726	0.819 ± 0.002	0.818 ± 0.003	-0.001	mlp_256 (528,897)	0.525	0.527	within seed spread
TextVQA	Verifier head	0.726	0.866 ± 0.002	0.869 ± 0.001	+0.003	mlp_256 (528,897)	0.639	0.647	head better
TextVQA	Unified, pre-generation mode	0.726	0.820 ± 0.001	0.814 ± 0.001	-0.006	mlp_512 (2,106,881)	0.533	0.511	linear probe better
TextVQA	Unified, joint mode	0.726	0.870 ± 0.000	0.867 ± 0.002	-0.002	mlp_512 (2,106,881)	0.642	0.635	linear probe better
DocVQA	Pre-generation head	0.685	0.773 ± 0.007	0.781 ± 0.004	+0.008	mlp_256 (528,897)	0.394	0.391	head better
DocVQA	Verifier head	0.685	0.788 ± 0.002	0.790 ± 0.000	+0.002	mlp_512 (1,053,697)	0.443	0.451	head better
DocVQA	Unified, pre-generation mode	0.685	0.766 ± 0.003	0.771 ± 0.005	+0.005	mlp_256 (1,057,537)	0.380	0.387	within seed spread
DocVQA	Unified, joint mode	0.685	0.806 ± 0.002	0.808 ± 0.002	+0.002	mlp_256 (1,057,537)	0.452	0.459	within seed spread
ScienceQA	Pre-generation head	0.654	0.901 ± 0.001	0.910 ± 0.007	+0.009	mlp_256 (528,897)	0.540	0.541	head better
ScienceQA	Verifier head	0.654	0.875 ± 0.006	0.878 ± 0.001	+0.004	mlp_256 (528,897)	0.486	0.505	within seed spread
ScienceQA	Unified, pre-generation mode	0.654	0.902 ± 0.003	0.902 ± 0.006	+0.000	mlp_512 (2,106,881)	0.539	0.537	within seed spread
ScienceQA	Unified, joint mode	0.654	0.897 ± 0.003	0.906 ± 0.006	+0.009	mlp_512 (2,106,881)	0.540	0.545	head better

> **READING THE COMPARISON**
> The trained heads are better than linear probes, but only a little. Trained on the pool, the head is ahead in all 16 comparisons by 0.001 to 0.013 AUROC (mean 0.006), and the lead is larger than the seed spread in 12 of them; the largest leads are the question-only head on DocVQA (+0.013), the verifier on ChartQA (+0.010) and the question-only head on TextVQA (+0.010). Trained on each dataset alone, with 6,000 to 28,000 training rows, the mean lead falls to 0.003: the head wins by more than the seed spread in 7 comparisons, ties in 7 (in two of them the best head is a linear probe, the unified head on ChartQA), and loses in 2, both for the unified head on TextVQA (-0.002 and -0.006). A larger head therefore needs the pooled data to pay off consistently. Either way the linear probe delivers about 90% to 99% of the head's gain over the answer-token probability (above 100% where the probe is better) with well under 1% of the parameters of the larger head, which makes it the sensible default and the larger heads a small refinement.

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9. Loss functions

Three loss designs were run on the same heads, features and selection protocol, so differences can be attributed to the loss.

9.1 The old loss

The old loss asks only for the right label on real inputs (Sections 4 and 7): cross-entropy, with a pre-generation term for the unified head. A head trained this way can reach a high AUROC by learning which kinds of questions the model tends to get wrong, without ever using what the image shows. That is useful for ranking, but it is not what a correction step needs, which is a score that reacts when the visual evidence is missing.

9.2 The new loss

The proposed loss has three lines, and each maps to one term:

$$P_\theta(S_{\text{conf}} \mid h_{\text{answer}}, h_{\text{question}}) \;\rightarrow\; L_{\text{cls}} \;(\text{and } L_{\text{pre}})$$

$$\mathcal{L}_{\text{contrastive}}(h_{\text{predicted}}, h_{\text{correct}}) \;\rightarrow\; L_{\text{pair}}$$

$$\mathcal{L}_{\text{contrastive}}(h_{\text{clean}}, h_{\text{corrupt}}) \rightarrow S_{\text{conf}} \;(\text{visual grounding}) \;\rightarrow\; L_{\text{ground}}$$

Written out for a mini-batch:

$$L = L_{\text{cls}} + w_{\text{pre}} L_{\text{pre}} + w_{\text{pair}} L_{\text{pair}} + \lambda\, L_{\text{ground}}$$

$$L_{\text{ground}} = \operatorname*{mean}_{m \in \text{modes}} \operatorname*{mean}_{i \in \text{correct}} \operatorname{softplus}\big(\text{margin} - (s_m(x_i^{\text{clean}}) - s_m(x_i^{\text{corrupt}}))\big), \qquad \text{margin} = 0$$

$$L_{\text{pair}} = \underset{i \in \text{wrong}}{\text{mean}}\ \text{softplus}\big(-(s(x_i^{\text{ref}}) - s(x_i^{\text{pred}}))\big)$$

Here x^{clean} is the real image with the question and the model's answer, x^{corrupt} is the same question and answer text with a corrupted image, x^{ref} appends the correct answer and x^{pred} appends the model's own answer. The old loss is the case $\lambda = 0$. The grounding term is applied only to answers the model got right: for those, the evidence is genuinely in the image, so removing the image should lower the score. For wrong answers there is nothing to ground, so they enter only through the correctness loss. The second line (the pair term) is our reading of the proposal's second contrastive term; an embedding-level contrastive loss would need a change to the head loss and was not tried.

9.3 Corrupting the image: swap and noise

The grounding term needs a corrupted image. Two were built, in this order.

- **Swap.** Replace the image with a different example's image from the same split. Simple and strong, but it has a weakness: the corrupted pair is also a mismatched question and image, so a head can satisfy the term by detecting the mismatch, without judging how well the image supports the answer.
- **Noise.** Keep the right image and mix Gaussian noise into it, the forward step of a diffusion process (Ho et al. 2020): $x' = \sqrt{\alpha}\, x + \sqrt{1-\alpha}\, \varepsilon$, $\varepsilon \sim \mathcal{N}(0, I)$, with $\alpha = 0.15$. The scene is still there but its detail is not, so there is no mismatch to detect. The strength was chosen from a smoke test of 30 examples per dataset, by how far the model's own answer probability falls.

Mask (60% of a 16 by 16 grid blanked) and patch shuffle (a 4 by 4 grid permuted) were also built. They were extracted for the test splits and used only to grade heads on corruptions they never trained on. Everything was run through the same pipeline: a pooled sweep for each corruption, a re-grade of the swap heads under the other corruptions, one sweep per dataset alone for each corruption, and sequential training (Section 10). Details: reports 05 (swap) and 06 (noise).

9.4 How grounding is measured

For each correct test answer, the head scores the pair with the clean image and with the corrupted one. **Order accuracy** is the share of correct answers whose score falls under corruption (0.5 means no effect, 1.0 means it always falls). It is the number quoted in the results as "the score falls". The untrained answer-token probability can be graded the same way.

9.5 Results: pooled mixture

Head	AUROC, old loss	AUROC, new loss (swap)	AUROC, new loss (noise)	Falls under noise, old loss	Falls under swap, new loss (swap)	Falls under noise, new loss (noise)
Separate pre-generation head	0.8643	0.8625	0.8625	0.921	0.918	0.971

Head	AUROC, old loss	AUROC, new loss (swap)	AUROC, new loss (noise)	Falls under noise, old loss	Falls under swap, new loss (swap)	Falls under noise, new loss (noise)
Separate verifier head	0.8727	0.8714	0.8731	0.941	0.986	1.000
Unified head	0.8813	0.8822	0.8837	0.921	0.963	0.991

The old loss column is the same head in every version. Differences between the loss columns in AUROC are no more than a few thousandths, inside the seed spread. Grounding is graded under noise for the old loss and for the noise version, and under swap for the swap version, as that is the corruption each head trained against.

9.6 What each term does (2 by 2 ablation, swap)

The two contrastive terms were switched on and off independently, keeping everything else fixed.

Head	Terms on	AUROC	Share whose score falls (swap)	AUROC change vs plain	Grounding change vs plain
Separate verifier	Neither term (plain)	0.873 ± 0.001	0.949 ± 0.001	+0.000	+0.000
Separate verifier	Pair term only	0.853 ± 0.005	0.902 ± 0.002	-0.019	-0.047
Separate verifier	Grounding term only	0.871 ± 0.001	0.986 ± 0.000	-0.001	+0.037
Separate verifier	Both terms	0.852 ± 0.002	0.969 ± 0.003	-0.020	+0.020
Unified, joint mode	Neither term (plain)	0.881 ± 0.002	0.931 ± 0.006	+0.000	+0.000
Unified, joint mode	Pair term only	0.870 ± 0.000	0.861 ± 0.002	-0.011	-0.070
Unified, joint mode	Grounding term only	0.882 ± 0.000	0.963 ± 0.002	+0.001	+0.032
Unified, joint mode	Both terms	0.872 ± 0.000	0.944 ± 0.002	-0.009	+0.013

> **IMPORTANT**
> The grounding term is nearly free: it changes AUROC by at most 0.001 in either direction (unified +0.001, verifier -0.001) and raises the share of correct answers whose score falls under a corrupted image by 0.03 to 0.04. The pair term is not free: on its own it lowers AUROC by 0.01 to 0.02. The loss as first specified, with both terms, is therefore slightly worse than the old loss in ranking, and the grounding term alone is the version worth keeping.

9.7 Does grounding learned on one corruption carry over?

Each row is graded on the same test rows under four corruptions. The mask and shuffle columns are corruptions never trained on.

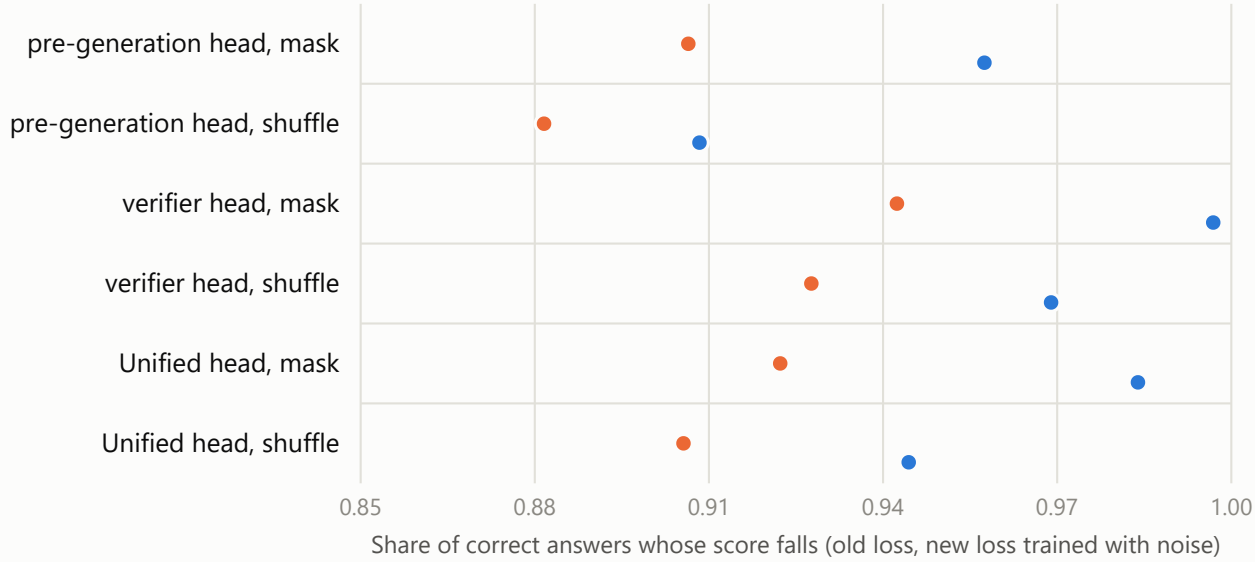
Separate verifier head

Trained with	Swap	Noise	Mask	Shuffle
Old loss	0.949	0.941	0.942	0.928
New loss, trained with swap	0.986	0.980	0.980	0.952
New loss, trained with noise	0.967	1.000	0.997	0.969

Unified head

Trained with	Swap	Noise	Mask	Shuffle
Old loss	0.931	0.921	0.922	0.906
New loss, trained with swap	0.963	0.950	0.952	0.922
New loss, trained with noise	0.952	0.991	0.984	0.944

Grounding on corruptions that were never trained on



Orange is the old loss and blue the noise-trained new loss. Both are graded on mask and shuffle.

Training with either corruption raises grounding on every other corruption compared with the old loss. For heads trained with swap the gain is about 0.03 to 0.04 on noise and mask and 0.015 to 0.024 on shuffle. For heads trained with noise the gain is largest on mask (0.05 to 0.06), which is expected because both remove detail instead of changing content. The gain is not specific to one corruption, although all four still alter the hidden state in ways a head can read.

9.8 Each dataset alone

Dataset	AUROC, old	AUROC, new (swap)	AUROC, new (noise)	Falls under noise, old	Falls under swap, new (swap)	Falls under noise, new (noise)
ChartQA	0.898	0.897	0.896	0.989	0.985	1.000
TextVQA	0.867	0.868	0.866	0.990	0.994	0.999
DocVQA	0.808	0.810	0.812	0.996	0.999	1.000
ScienceQA	0.906	0.901	0.905	0.722	0.881	0.938

On ChartQA, TextVQA and DocVQA the old-loss head already lowers its score on about 0.97 or more of correct answers under noise (all three head kinds), so there is little room and AUROC does not move. ScienceQA is where the term matters: the old loss lowers the score on 0.69 to 0.78 of correct answers under noise across the three head kinds, close to indifferent, and the noise loss raises that to 0.87 to 1.0 (the table shows the unified head: 0.72 to 0.94). ScienceQA questions can often be answered from the text alone, so a score that ignores the image is the expected outcome without the term.

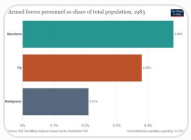
> **HOW FAR TO TRUST THE GROUNDING NUMBERS**
> Order accuracy near 1.0 is partly easy. A noised image moves the hidden state a lot, so a head can learn to lower its score on noisy-looking inputs without checking whether the image supports the answer. The cross-corruption check lowers this worry but does not remove it. A stronger test is a corruption that removes only the evidence for the answer (for example, masking only the region the answer comes from), which was not run. The noise strength was set on a 30-example smoke test and only one strength was trained.

9.9 What the corrupted images look like

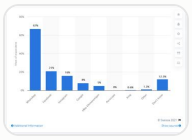
The same clean test image, the swapped partner (another example's image from the same split), and the noised version used by the noise loss. The noised image keeps the layout but loses most of the detail, which is why a head cannot satisfy the noise term by detecting a question and image mismatch. Note that for ScienceQA the swapped partner is another map of a similar kind, so the swap is a milder corruption there than on the other datasets, which may be part of why grounding is lowest on ScienceQA with swap.

ChartQA

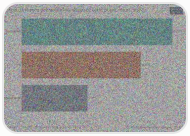
Clean



Swapped



Noised (alpha 0.15)



TextVQA

Clean

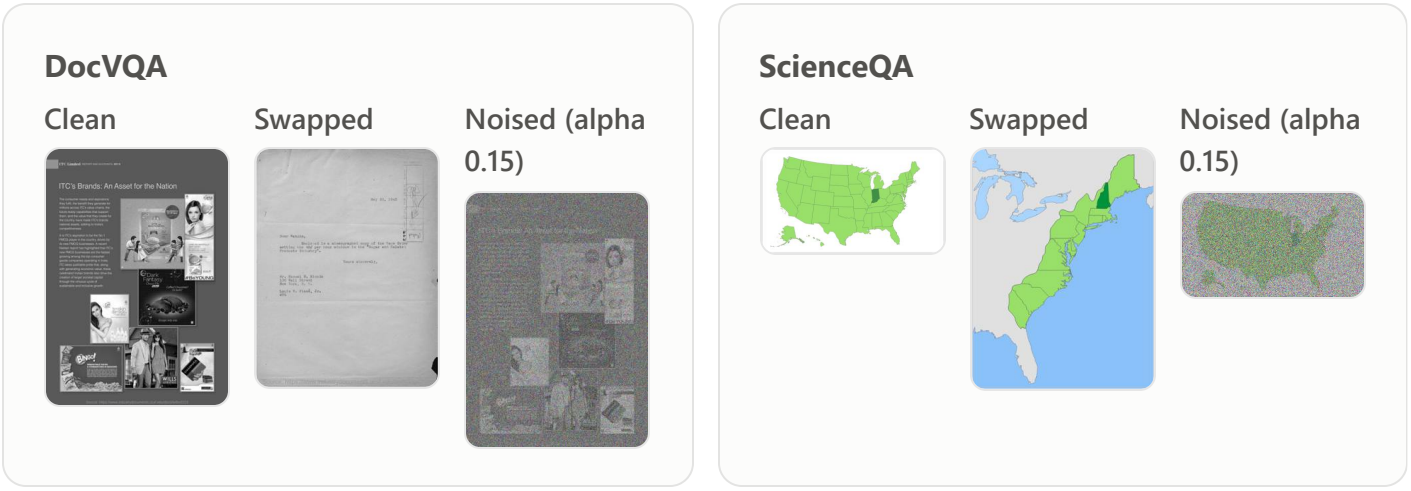


Swapped



Noised (alpha 0.15)





9.10 The grounding weight

The grounding weight λ trades ranking quality for grounding. Mean validation AUROC over all configurations that use each value of λ (swap and noise sweeps):

Corruption	Head	λ	Configurations	Mean validation AUROC	Best validation AUROC
swap	Separate verifier head	0	12	0.8388	0.8573
swap	Separate verifier head	0.1	12	0.8369	0.8546
swap	Separate verifier head	0.5	12	0.8318	0.8502
swap	Unified head	0	24	0.8604	0.8711
swap	Unified head	0.1	24	0.8615	0.8711
swap	Unified head	0.5	24	0.8595	0.8688
noise	Separate verifier head	0	12	0.8388	0.8573
noise	Separate verifier head	0.1	12	0.8363	0.8569
noise	Separate verifier head	0.5	12	0.8330	0.8537
noise	Unified head	0	24	0.8604	0.8711
noise	Unified head	0.1	24	0.8591	0.8730
noise	Unified head	0.5	24	0.8549	0.8679

Going from $\lambda = 0$ to 0.1 changes mean validation AUROC by 0.003 or less (up for the swap unified head, down for the others); going to 0.5 costs 0.001 to 0.007. The sweep therefore picks 0.1, and the grounding gains reported in this document are at that setting.

9.11 All head kinds on each dataset alone

The same comparison as in Section 9.8 for the pre-generation head, the verifier and the unified head.

Dataset	Head	AUROC, old	AUROC, new (swap)	AUROC, new (noise)	Falls under noise, old	Falls under swap, new (swap)	Falls under noise, new (noise)
ChartQA	Separate pre-generation head	0.896	0.896	0.896	0.981	0.949	0.998
ChartQA	Separate verifier head	0.894	0.897	0.893	0.969	0.998	1.000
ChartQA	Unified head	0.898	0.897	0.896	0.989	0.985	1.000
TextVQA	Separate pre-generation head	0.818	0.815	0.813	0.972	0.939	0.994
TextVQA	Separate verifier head	0.869	0.868	0.868	0.990	0.995	0.999
TextVQA	Unified head	0.867	0.868	0.866	0.990	0.994	0.999
DocVQA	Separate pre-generation head	0.781	0.784	0.784	0.984	0.966	0.999
DocVQA	Separate verifier head	0.790	0.791	0.791	0.991	0.998	1.000
DocVQA	Unified head	0.808	0.810	0.812	0.996	0.999	1.000
ScienceQA	Separate pre-generation head	0.910	0.906	0.901	0.685	0.846	0.872
ScienceQA	Separate verifier head	0.878	0.880	0.884	0.781	0.925	0.999
ScienceQA	Unified head	0.906	0.901	0.905	0.722	0.881	0.938

10. Stage 6: sequential training and forgetting

The question. The mixture trains on all datasets at once, which is the easy case. In practice a head may be trained on dataset A, then updated on B and C. Does it still score A well afterwards? This is catastrophic forgetting, and it decides whether the heads can be extended to new data without retraining from scratch.

10.1 Protocol

One head is trained through the four datasets in a fixed order, continuing from the previous weights, with early stopping on each stage's validation split. After every stage the head is scored on every dataset's test split, giving a stage-by-dataset matrix of AUROC. Three orders (forward, reverse, shuffled) and three seeds are averaged. The head configurations are the winners of the pooled sweeps, so the old and new loss are compared at the settings chosen on validation. Two variants of each loss were run: plain sequential training, and with a replay buffer, which mixes a fixed 10% random sample of every earlier dataset's training rows into each later stage.

Let $M_{t,d}$ be the AUROC on dataset d after stage t, T the number of stages, and $\tau(d)$ the stage at which d is trained. The metrics are:

$$\text{final average} = \frac{1}{D}\sum_{d} M_{T,d}, \qquad \text{forgetting} = \frac{1}{D-1}\sum_{d \neq \text{last}} \Big(\max_{t \geq \tau(d)} M_{t,d} - M_{T,d} \Big)$$

$$\text{backward transfer} = \frac{1}{D-1}\sum_{d \neq \text{last}} \big(M_{T,d} - M_{\tau(d),d} \big), \qquad \text{retention} = \frac{1}{D}\sum_{d} \big(M_{T,d} - M_{d}^{\text{alone}} \big)$$

Retention compares the final head with a head trained on that dataset alone. The last column of the tables is the grounding order accuracy after the final stage, averaged over datasets.

10.2 Results with the swap loss

Head	Variant	Final average AUROC	Forgetting	Backward transfer	Retention	Grounding after last stage
Separate pre-generation head	Old loss, no replay	0.839	0.013	-0.012	-0.005	0.856
Separate pre-generation head	Old loss, 10% replay	0.841	0.013	-0.011	-0.003	0.856
Separate pre-generation head	New loss, no replay	0.833	0.020	-0.020	-0.015	0.914
Separate pre-generation head	New loss, 10% replay	0.837	0.019	-0.019	-0.011	0.925
Separate verifier head	Old loss, no replay	0.834	0.032	-0.032	-0.021	0.933
Separate verifier head	Old loss, 10% replay	0.856	0.005	-0.005	-0.000	0.938
Separate verifier head	New loss, no replay	0.832	0.034	-0.034	-0.024	0.959
Separate verifier head	New loss, 10% replay	0.854	0.006	-0.006	-0.002	0.979
Unified head	Old loss, no replay	0.865	0.011	-0.011	-0.005	0.924
Unified head	Old loss, 10% replay	0.865	0.013	-0.012	-0.005	0.921
Unified head	New loss, no replay	0.863	0.013	-0.013	-0.006	0.951
Unified head	New loss, 10% replay	0.864	0.015	-0.014	-0.005	0.963

10.3 Results with the noise loss

Head	Variant	Final average AUROC	Forgetting	Backward transfer	Retention	Grounding after last stage
Separate pre-generation head	Old loss, no replay	0.839	0.013	-0.012	-0.005	0.909
Separate pre-generation head	Old loss, 10% replay	0.841	0.013	-0.011	-0.003	0.908
Separate pre-generation head	New loss, no replay	0.828	0.025	-0.024	-0.018	0.970
Separate pre-generation head	New loss, 10% replay	0.837	0.016	-0.016	-0.009	0.980
Separate verifier head	Old loss, no replay	0.834	0.033	-0.033	-0.022	0.904
Separate verifier head	Old loss, 10% replay	0.855	0.006	-0.005	-0.001	0.918
Separate verifier head	New loss, no replay	0.823	0.046	-0.046	-0.036	0.992
Separate verifier head	New loss, 10% replay	0.853	0.008	-0.007	-0.005	1.000
Unified head	Old loss, no replay	0.865	0.011	-0.011	-0.005	0.917
Unified head	Old loss, 10% replay	0.865	0.013	-0.012	-0.005	0.914
Unified head	New loss, no replay	0.863	0.013	-0.013	-0.007	0.969
Unified head	New loss, 10% replay	0.864	0.014	-0.013	-0.005	0.986

Unified head, old loss, forward order: AUROC on every dataset after every stage

	ChartQA	TextVQA	DocVQA	ScienceQA
After ChartQA	0.901	0.804	0.766	0.873
After TextVQA	0.884	0.873	0.803	0.878
After DocVQA	0.885	0.864	0.816	0.858
After ScienceQA	0.880	0.863	0.808	0.908

Rows are stages, columns are the datasets scored. The cells below and to the left of the diagonal show datasets seen earlier.

Unified head, new loss, forward order: AUROC on every dataset after every stage

	ChartQA	TextVQA	DocVQA	ScienceQA
After ChartQA	0.899	0.811	0.769	0.861
After TextVQA	0.883	0.873	0.804	0.873
After DocVQA	0.883	0.864	0.816	0.854
After ScienceQA	0.880	0.863	0.804	0.909

Rows are stages, columns are the datasets scored. The cells below and to the left of the diagonal show datasets seen earlier.

10.4 Heads trained on one dataset alone, for reference

Dataset	Unified joint AUROC when trained alone
ChartQA	0.901 ± 0.001
TextVQA	0.872 ± 0.001
DocVQA	0.810 ± 0.001
ScienceQA	0.897 ± 0.012

FINDINGS

Heads trained through all four datasets in sequence keep most of what they learned: forgetting is about 0.011 to 0.015 AUROC for the unified head, 0.013 to 0.025 for the pre-generation head, and the answer-conditioned verifier forgets the most (0.032 to 0.046 without replay). A 10% replay buffer cuts the verifier's forgetting to about 0.005 to 0.008 and changes little for the others. The new loss does not change forgetting much for the unified head, and the noise loss is slightly worse than the swap loss without replay on the pre-generation and verifier heads, with replay removing the difference. Grounding is not forgotten: after the last stage the share of correct answers whose score falls is higher with the new loss than with the old loss in every row.

10.5 The other dataset orders

The forward order (ChartQA, TextVQA, DocVQA, ScienceQA) puts the largest dataset first. Two further orders let you see whether the pattern depends on it: reverse (ScienceQA first) and a shuffled order. The tables above average the three orders; here are the stage matrices for the unified head in the other two.

Unified head, old loss, reverse order

	ChartQA	TextVQA	DocVQA	ScienceQA
After ScienceQA	0.809	0.617	0.653	0.897
After DocVQA	0.869	0.822	0.811	0.873
After TextVQA	0.861	0.875	0.810	0.891
After ChartQA	0.900	0.846	0.800	0.894

Unified head, new loss, reverse order

	ChartQA	TextVQA	DocVQA	ScienceQA
After ScienceQA	0.825	0.662	0.678	0.896
After DocVQA	0.869	0.826	0.811	0.869
After TextVQA	0.859	0.874	0.806	0.888
After ChartQA	0.901	0.845	0.796	0.891

Unified head, old loss, shuffled order

	ChartQA	TextVQA	DocVQA	ScienceQA
After DocVQA	0.866	0.808	0.810	0.763
After ScienceQA	0.866	0.810	0.802	0.910
After ChartQA	0.898	0.822	0.794	0.907
After TextVQA	0.888	0.875	0.810	0.905

Unified head, new loss, shuffled order

	ChartQA	TextVQA	DocVQA	ScienceQA
After DocVQA	0.867	0.817	0.812	0.759
After ScienceQA	0.871	0.823	0.806	0.906
After ChartQA	0.900	0.821	0.793	0.904
After TextVQA	0.887	0.875	0.809	0.902

10.6 What each dataset loses, one by one

For the unified head in the forward order: the AUROC on each earlier dataset right after its own stage, at the end, and the drop from the best score it had after its own stage. The last dataset is not listed because nothing came after it.

Variant	Dataset	AUROC after its own stage	AUROC at the end	Drop from its best
Old loss	ChartQA	0.901	0.880	+0.021
Old loss	TextVQA	0.873	0.863	+0.010
Old loss	DocVQA	0.816	0.808	+0.008
New loss	ChartQA	0.899	0.880	+0.019
New loss	TextVQA	0.873	0.863	+0.010
New loss	DocVQA	0.816	0.804	+0.013
Old loss, 10% replay	ChartQA	0.901	0.876	+0.025
Old loss, 10% replay	TextVQA	0.871	0.846	+0.025
Old loss, 10% replay	DocVQA	0.816	0.778	+0.038
New loss, 10% replay	ChartQA	0.899	0.878	+0.022
New loss, 10% replay	TextVQA	0.872	0.843	+0.029
New loss, 10% replay	DocVQA	0.817	0.780	+0.037

Without replay the drops are 0.008 to 0.021 and are largest for the dataset trained first (ChartQA), which has been through the most later stages. With a replay buffer the unified head's drops in this order are larger (0.022 to 0.038), so replay does not help the unified head here, which matches the sequential tables in Section 10.2 where

replay leaves its forgetting unchanged or slightly higher. Replay helps the separate verifier head, not the unified head.

---

11. The trained heads

Every head that mattered for a result is saved in the repository under `trained_heads/approach1/`, with a loading guide in `trained_heads/README.md`. The sweeps kept weights only in memory, so the winning configurations were retrained with the same data, seeds and early stopping and checked against the sweep results.

- **468 sweep winners** (156 configurations on three seeds): every winning configuration for the pooled mixture, the noise mixture, and each dataset alone with swap and with noise, for all head kinds and for the old loss, new loss, linear probes and the four ablation cells.
- **72 sequential heads**: the final head of each sequential variant and order (seed 1), for the swap and noise mixtures.

Folder	Trained on	Configurations	Files	Largest gap to sweep mean
winners/mixture	Pooled mixture, swap	16	48	0.0006
winners/mixture_noise	Pooled mixture, noise	16	48	0.0000
winners/chartqa_swap	ChartQA alone, swap	14	42	0.0047
winners/chartqa_noise	ChartQA alone, noise	14	42	0.0001
winners/textvqa_swap	TextVQA alone, swap	16	48	0.0015
winners/textvqa_noise	TextVQA alone, noise	16	48	0.0000
winners/docvqa_swap	DocVQA alone, swap	16	48	0.0000
winners/docvqa_noise	DocVQA alone, noise	16	48	0.0000
winners/scienceqa_swap	ScienceQA alone, swap	16	48	0.0023
winners/scienceqa_noise	ScienceQA alone, noise	16	48	0.0000

The last column is the largest gap between the three-seed mean test AUROC of the saved heads and the sweep's mean. The pooled and noise folders reproduce to under 0.001, and the largest gap anywhere is 0.005, about the seed spread. The Qwen3-VL-2B model itself is not included (it is never changed) and the cached features are regenerated with `extract_q_features.py`.

```
from pathlib import Path
from experiments.approach1_latent_self_correction.export_heads import load_head, score
head, spec, temperatures =
load_head(Path("trained_heads/approach1/winners/mixture_noise/unified_new_seed1.pt"))
probabilities = score(head, spec, temperatures, prompt_state, answer_state)   # {"joint":
..., "pre_generation": ...}
```

These exported heads are the ones the correction experiments of Section 12 load, which shows they work as a component and not only as a table of results.

12. Stage 8: using the head to correct answers

The previous sections built the gate. These experiments test whether acting on it helps. Sections 12.1 to 12.5 are the pilot, run on 100 examples; Section 12.6 repeats everything on the full test split and supersedes the pilot numbers. All of them use the frozen Qwen3-VL-2B and the exported pooled-mixture unified head (`trained_heads/approach1/winners/mixture/unified_new_seed1.pt`), on a balanced study set of 100 examples, 25 from each dataset, of the pooled test split. These experiments were pushed to the `feature/latent-self-correction` branch (commit `e6aaa80`) and are not on `main` yet. The code is in `prompted_retry.py`, `online_prompted_retry.py`, `candidate_reranking.py` and `deepen_selected_candidate.py`, run by `run_prompted_retry.sbatch`, `run_candidate_reranking.sbatch` and `run_deepen_selected.sbatch`, with sampling options added to `models/run_inference.py` and tests in `tests/test_prompted_retry.py`. The full report is 07. Accuracy is the dataset's own metric converted to the binary correctness definition used to train the head.

> **ABOUT "CHAIN OF THOUGHT"**
> The deeper pass (Section 12.4) is the experiment described as chain of thought: after a candidate is chosen, the model is told to *think more deeply*, re-examine the visual evidence and correct the candidate. It ran on the same 100 examples with a limit of 64 new tokens. The prompt still ends with *respond with only the final answer, without explanation*, and most of the time the model complied with a short answer. It did not always. In 5 cases (all ChartQA) it wrote out reasoning anyway, such as a list of the chart values or a subtraction, and ran into the 64-token limit before reaching a final answer. In 5 further cases (4 in DocVQA, 1 in TextVQA) it repeated the prompt text instead of answering. So the run mixes direct answers, a few cut-off chains of thought and a few format failures. It is not a controlled chain-of-thought experiment: reasoning was not requested, was not given enough tokens, and no final answer was extracted from it. A proper run (ask for reasoning, allow enough tokens, extract the final answer) has not been done.

12.1 Prompted retry

Idea. The model answers once. A second prompt shows it the previous answer and asks it to inspect the visual evidence again. The head decides *when* to retry, so the retry's cost and its risk of overturning correct answers are spent only where the gate thinks the first answer is weak. Each policy is evaluated on the same retry outputs, so there is no generation noise between rows. A random 20% control has the same compute as the head-targeted policies.

```
Your previous answer was: {answer}
Try the question again. Carefully inspect the visual evidence, including labels, values,
and spatial relationships. Do not assume the previous answer is correct.
Respond with only the final answer, without explanation.
```

Policy	Accuracy	Retried	Fixed	Broken	ChartQA	TextVQA	DocVQA	ScienceQA
No retry (greedy answer kept)	81.0%	0	+0	-0	56%	88%	92%	88%
Retry every answer	78.0%	100	+4	-7	60%	88%	88%	76%

Policy	Accuracy	Retried	Fixed	Broken	ChartQA	TextVQA	DocVQA	ScienceQA
Retry if head score below 0.5	78.0%	17	+2	-5	56%	88%	92%	76%
Retry the 20% with the lowest head score	78.0%	20	+2	-5	56%	88%	92%	76%
Retry the 20% with the lowest answer-token probability	76.0%	20	+1	-6	52%	88%	88%	76%
Retry a random 20% (control)	82.0%	20	+1	-0	56%	88%	92%	92%

This first run began at 81% for the greedy answer. Retrying every answer lowered accuracy to 78%, and head-based targeting also reached 78%, while the random control reached 82%. In this run a single prompted retry is unstable: it fixes some errors but overturns more correct answers, and the head's flagging did not beat random choice.

12.2 Scored retry and reranking

Idea. Instead of trusting the retry, score both the original and the retry with the same head and keep the one with the higher score. Gated variants also require the original to score below 0.5 and optionally require a minimum improvement, so that the retry replaces the original only with good evidence.

Policy	Accuracy	Retried	Fixed	Broken	ChartQA	TextVQA	DocVQA	ScienceQA
No retry (greedy answer kept)	79.0%	0	+0	-0	52%	84%	92%	88%
Retry every answer	79.0%	100	+4	-4	60%	84%	92%	80%
Retry if head score below 0.5	80.0%	18	+3	-2	56%	84%	92%	88%
Retry the 20% with the lowest head score	80.0%	20	+3	-2	56%	84%	92%	88%
Retry the 20% with the lowest answer-token probability	78.0%	20	+1	-2	52%	84%	92%	84%
Retry a random 20% (control)	78.0%	20	+0	-1	52%	84%	92%	84%
Head reranking: keep the higher-scored of original and retry	79.0%	4	+2	-2	56%	84%	92%	84%
Gated: retry only if original below 0.5 and retry scores higher	80.0%	3	+2	-1	56%	84%	92%	88%
Gated, margin 0.05	79.0%	2	+1	-1	52%	84%	92%	88%
Gated, margin 0.10	79.0%	0	+0	-0	52%	84%	92%	88%
Gated, margin 0.20	79.0%	0	+0	-0	52%	84%	92%	88%
Oracle: use the retry exactly when it fixes an error	83.0%	4	+4	-0	60%	84%	92%	96%

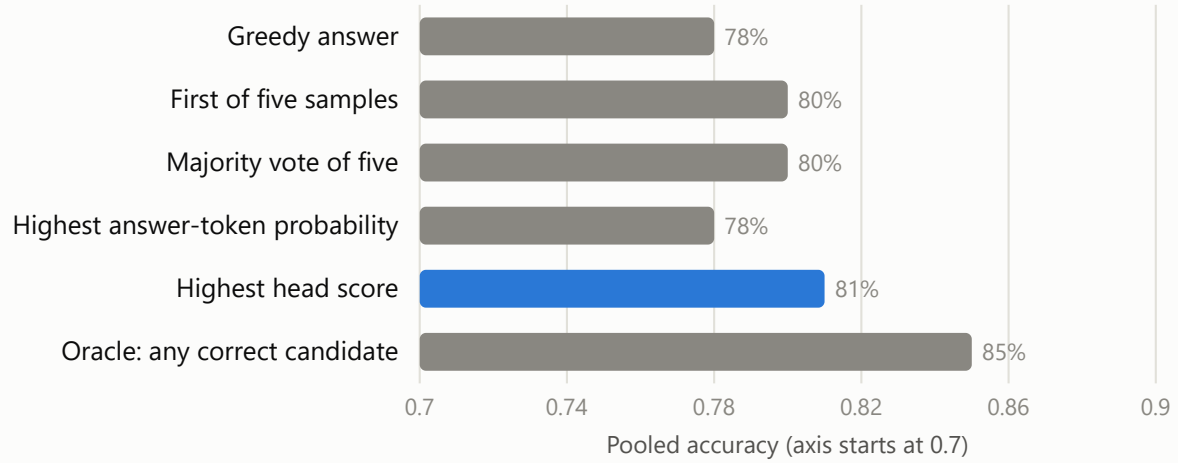
This run was generated independently and its greedy baseline is 79%. Here the head-targeted retries (below 0.5, or the lowest 20%) reached 80% with 3 fixes and 2 breaks, above the random control (78%) and the baseline (79%), and the gated reranking also reached 80% (2 fixes, 1 break). The oracle is 83%. So in this run the head-guided retry is one point better than doing nothing, and in the first run it was three points worse than doing nothing. The two runs differ because of generation and GPU nondeterminism, and the effect in both is within a few questions out of 100, so the honest reading is that a prompted retry does not clearly help, with or without the head. Comparisons are only meaningful within a table.

12.3 Best-of-five candidate selection

Idea. Separate creating candidates from choosing between them. Five answers are sampled at temperature 0.9 and top-p 0.95. The head scores every candidate from its answer-state h_a, and the highest score is selected: $\hat{a} = \arg\max_k \; p_\theta(\text{correct} \mid h_q, h_{a_k})$. The comparison methods are the first sample, a majority vote over normalized answer text, the sample with the highest answer-token probability, the greedy baseline, and an oracle that picks a correct candidate whenever one exists (an unattainable upper bound).

Selection method	Accuracy	95% interval	ChartQA	TextVQA	DocVQA	ScienceQA
Greedy answer	78.0%	69% to 85%	52%	84%	92%	84%
First of five samples	80.0%	71% to 87%	52%	88%	92%	88%
Majority vote of five	80.0%	71% to 87%	52%	88%	92%	88%
Highest answer-token probability	78.0%	69% to 85%	52%	84%	92%	84%
Highest head score	81.0%	72% to 87%	52%	92%	96%	84%
Oracle: any correct candidate	85.0%	77% to 91%	56%	92%	96%	96%

Best-of-five selection on 100 examples



Blue is the head selector. With 100 examples, differences of a few points are within noise.

HOW STRONG IS THE EVIDENCE?

Each study has 100 examples. At that size, 81% accuracy has a 95% interval of about 72% to 87% and 78% has 69% to 85%, so the intervals overlap heavily. Head selection changed the outcome of 5 questions against greedy: it fixed 4 and broke 1. A paired sign test on 4 against 1 gives a two-sided p-value of about 0.38. The direction is encouraging, but the experiment is too small to claim the improvement is real. A run on the full test split is the next step. The candidate-level check is cleaner: over the 500 sampled answers, the head's score has AUROC 0.879 for telling correct from wrong candidates, against 0.813 for the answer-token probability.

Cases where head selection changed the outcome

Every example where the head-selected candidate was right and greedy was wrong, or the reverse. Long ScienceQA contexts are shortened.

Example	Question	Reference	Greedy answer	Head-selected candidate
textvqa:21	what number is the player on the right?	9	3 (wrong)	9 (head 0.914, correct)
textvqa:22	what football league is the jacket from on the man pointing?	ryman	ryman football league (wrong)	ryman (head 0.691, correct)
docvqa:0	What is the 'actual' value per 1000, during the year 1975?	0.28	0.27 (wrong)	0.28 (head 0.771, correct)
scienceqa_img:7	Context: A boy plays with marbles. He pushes one of the marbles with his thumb. Question: What is the direction of this push? Options: A. toward th...	B	A (wrong)	B (head 0.367, correct)
scienceqa_img:13	Context: The map below shows air temperatures in the lower atmosphere on October 28, 2016. The outlined area shows an air mass that influenced weat...	A	A (correct)	B (head 0.636, wrong)

Candidate sets contained 1.36 unique answers on average and all five samples agreed on 76% of examples. Head selection improved greedy accuracy by three points and captured three of the seven points between greedy and the oracle. The head selector (81%) scored one point above the majority vote (80%) and three above the answer-token probability (78%). These are small margins on 100 examples. Note that the first sample alone (80%) is already close to the head's choice, so most of the gain over greedy may come from sampling and not from selection; only the larger run can separate the two.

12.4 A deeper pass after selection

Idea. After the head picks a candidate, the model sees that candidate and is asked to think more deeply and correct it. The head then decides whether the deeper answer replaces the candidate.

Policy	Accuracy	Fixed	Broken	ChartQA	TextVQA	DocVQA	ScienceQA
Keep the head-selected candidate	81.0%	+0	-0	52%	92%	96%	84%
Replace with the deeper answer, always	73.0%	+2	-10	48%	80%	76%	88%
Keep the higher head score of the two	81.0%	+0	-0	52%	92%	96%	84%
Deeper answer if its head score is higher	81.0%	+0	-0	52%	92%	96%	84%
Deeper answer if at least 0.05 higher	81.0%	+0	-0	52%	92%	96%	84%
Deeper answer if at least 0.10 higher	81.0%	+0	-0	52%	92%	96%	84%
Deeper answer if at least 0.20 higher	81.0%	+0	-0	52%	92%	96%	84%
Oracle: deeper answer exactly when it fixes the candidate	83.0%	+2	-0	52%	92%	96%	92%

Replacing every selected candidate with the deeper answer lowered accuracy from 81% to 73% (2 fixed, 10 broken). Every head-based replacement rule left the answers unchanged: the head never scored the deeper

answer high enough to replace the selected candidate. The oracle for this step is 83%, so there is only a small upper bound to gain even with a perfect rule.

Is the drop only a format problem? Partly. Of the 10 answers the deeper pass broke, 5 are format failures: 1 was cut off at the 64-token limit and 4 repeated the prompt instead of answering. The other 5 are real changes of a correct answer to a wrong one, against 2 real fixes. So fixing the format would shrink the loss but would not turn it into a gain on this sample, and a controlled run with a larger token budget and answer extraction is needed before concluding anything about chain of thought.

12.5 What the pilot showed

> **FINDINGS**
> (1) The trained head works as a component: loaded from the repository and used to score freshly generated answers, it gives a candidate-level AUROC of 0.879 against 0.813 for the answer-token probability. (2) Asking the same model to retry or think more deeply is unreliable. It fixes some errors and breaks others. With this head as the gate the net effect ranged from -3 to +1 points between runs and policies, and the deeper pass was clearly harmful when applied to everything (-8 points). (3) Generating several candidates and letting the head choose looked promising in the pilot (81% against 78% greedy, 85% oracle). (4) All of this was on 100 examples, so none of the differences was statistically established, and the full-split rerun in Section 12.6 shows the best-of-five gain was mostly noise.

12.6 The full test split

The 100-example runs were a pilot. Every experiment above was rerun on the **full test splits of all four datasets: 9,517 questions** (ChartQA 2,500, TextVQA 2,500, DocVQA 2,500, ScienceQA 2,017), with the same prompts, sampling settings (temperature 0.9, top-p 0.95, five candidates), the same exported unified head, and the same 64-token limit. Each experiment ran as eight shards on separate GPUs and the shards were merged afterwards, so the policies that rank the whole set (the lowest 20% by head score, the random control) are computed once over all questions. There were no failed examples. Differences from a baseline are paired (the same questions under both policies) and come with a bootstrap 95% interval and an exact sign test (McNemar) on the questions where the two policies disagree. Accuracy differences are in percentage points.

> **RESULT IN ONE PARAGRAPH**
> On the full test split the 100-example optimism does not hold. Choosing among five samples with the head improves on greedy generation by only +0.40 points (95% interval -0.04 to +0.84, p = 0.081), which is not distinguishable from zero, against +3 points in the pilot. Prompted retry lowers accuracy unless it is tightly gated, where it gains at most +0.2 points. The deeper pass lowers accuracy by 12.8 points when applied to everything. The oracles show real headroom (+4.8 points for best-of-five, +2.9 for retry), so the head's scores rank candidates well (AUROC 0.883 against 0.815 for the answer-token probability over 47,585 sampled answers) but that ranking is not yet turned into accuracy.

Best-of-five candidate selection

Selection method	Accuracy	95% interval	Difference from greedy (95% interval)	Fixed / broken	p
Greedy answer	82.78%	82.0% to 83.5%			
First of five samples	81.97%	81.2% to 82.7%	-0.81 (-1.21 to -0.41)	+152 / -229	<0.001
Majority vote of five	82.86%	82.1% to 83.6%	+0.08 (-0.25 to +0.42)	+141 / -133	0.672
Highest answer-token probability	82.89%	82.1% to 83.6%	+0.12 (-0.09 to +0.32)	+60 / -49	0.338
Highest head score	83.18%	82.4% to 83.9%	+0.40 (-0.04 to +0.84)	+244 / -206	0.081
Oracle: any correct candidate	87.57%	86.9% to 88.2%	+4.79 (+4.35 to +5.24)	+470 / -14	<0.001

Differences are against the greedy answer; "fixed / broken" counts questions the policy got right where greedy was wrong and the reverse. Candidate sets contained 1.36 distinct answers on average. Head selection is 1.2 points better than taking the first sample ($p < 0.001$), so sampling alone does not help (the first sample is worse than greedy) and selection recovers that loss, but it does not beat greedy. The head is 0.3 points above majority vote, which is also not significant.

Dataset	Questions	Greedy	Head-selected	Oracle	Difference	Fixed / broken	p
ChartQA	2,500	76.6%	76.2%	82.4%	-0.40	+64 / -74	0.444
TextVQA	2,500	81.2%	82.3%	86.4%	+1.04	+86 / -60	0.038
DocVQA	2,500	87.0%	87.5%	90.6%	+0.48	+49 / -37	0.235
ScienceQA	2,017	87.1%	87.6%	91.7%	+0.50	+45 / -35	0.314

Only TextVQA shows a gain larger than noise (+1.0 points, $p = 0.04$, one of four datasets, so not corrected for multiple comparisons). On ChartQA, where greedy is weakest, head selection is slightly worse than greedy.

Prompted retry and reranking

Policy	Questions retried	Accuracy	Difference from no retry (95% interval)	Fixed / broken	p
No retry (greedy answer kept)	0	82.73%			
Retry every answer	9,517	79.99%	-2.73 (-3.33 to -2.15)	+272 / -532	<0.001
Retry if head score below 0.5	1,338	82.62%	-0.11 (-0.46 to +0.24)	+144 / -154	0.602
Retry the 20% with the lowest head score	1,904	82.27%	-0.45 (-0.88 to -0.03)	+205 / -248	0.048
Retry the 20% with the lowest answer-token probability	1,904	81.91%	-0.82 (-1.24 to -0.42)	+156 / -234	<0.001
Retry a random 20% (control)	1,904	82.05%	-0.67 (-0.92 to -0.43)	+47 / -111	<0.001
Head reranking: keep the higher-scored of original and retry	392	82.72%	-0.01 (-0.38 to +0.33)	+151 / -152	1.000
Gated: retry only if original below 0.5 and retry scores higher	237	82.97%	+0.24 (-0.03 to +0.50)	+100 / -77	0.098
Gated, margin 0.05	130	82.95%	+0.22 (+0.01 to +0.44)	+66 / -45	0.057

Policy	Questions retried	Accuracy	Difference from no retry (95% interval)	Fixed / broken	p
Gated, margin 0.10	56	82.84%	+0.12 (-0.03 to +0.26)	+30 / -19	0.152
Gated, margin 0.20	12	82.73%	+0.00 (-0.06 to +0.06)	+4 / -4	1.000
Oracle: use the retry exactly when it fixes an error	272	85.58%	+2.86 (+2.52 to +3.21)	+272 / -0	<0.001

Differences are against keeping the greedy answer. Retrying everything loses 2.7 points: the retry fixes 272 questions and breaks 532. Sending only the lowest 20% by head score to the retry loses less (0.45 points) than the random control (0.67) and the answer-token probability (0.82), so the head does point the retry at the right questions more often than chance, but the retry itself overturns too many correct answers for that to pay off. The gated policies (retry only if the original scores below 0.5 and the retry scores higher, optionally by a margin) are the only ones that help, by +0.24 points for margin 0 and about the same for 0.05, with intervals that touch zero.

Dataset	Greedy accuracy	Retry all	Lowest 20% by head	Random 20%	Gated, margin 0
ChartQA	76.6%	-1.84	-1.24	-0.36	-0.12
TextVQA	81.3%	-2.00	-0.76	-0.40	-0.04
DocVQA	87.0%	-0.92	-0.16	-0.32	+0.08
ScienceQA	86.8%	-6.99	+0.55	-1.83	+1.24

Change in accuracy by dataset, in points. ScienceQA is where retrying everything hurts most (-7.0), and where gating helps most (+1.2).

Deeper pass after best-of-five

Policy	Accuracy	Difference from the selected candidate (95% interval)	Fixed / broken	p
Keep the head-selected candidate	83.18%			
Replace with the deeper answer, always	70.39%	-12.79 (-13.53 to -12.08)	+90 / -1307	<0.001
Keep the higher head score of the two	83.17%	-0.01 (-0.16 to +0.14)	+25 / -26	1.000
Deeper answer if its head score is higher	83.17%	-0.01 (-0.16 to +0.14)	+25 / -26	1.000
Deeper answer if at least 0.05 higher	83.22%	+0.04 (-0.06 to +0.15)	+16 / -12	0.572
Deeper answer if at least 0.10 higher	83.28%	+0.11 (+0.01 to +0.20)	+16 / -6	0.052
Deeper answer if at least 0.20 higher	83.21%	+0.03 (+0.00 to +0.07)	+3 / -0	0.250
Oracle: deeper answer exactly when it fixes the candidate	84.12%	+0.95 (+0.76 to +1.16)	+90 / -0	<0.001

Applied to every selected answer, the deeper pass lowers accuracy from 83.2% to 70.4%: it fixes 90 questions and breaks 1307. A large share of that is format failure and not wrong reasoning: 526 of the 9,517 deeper answers hit the 64-token limit while writing out reasoning, and 552 repeated the prompt text. Those two kinds account for 751 of the 1307 breaks (57%); the other 556 are real changes of a correct answer to a wrong one. So the pass is clearly harmful as designed, and part of the harm could be removed by a larger token budget and answer

extraction, but the genuine breaks alone (556) still outnumber the fixes (90). Letting the head decide gives at most +0.1 points.

Pilot against full split

Comparison	Pilot, 100 examples	Full split, 9,517 questions
Best-of-five: head selection minus greedy	+3.0 points (n=100)	+0.40 points (n=9,517)
Best-of-five: oracle minus greedy	+7.0 points	+4.79 points
Prompted retry of everything	-3.0 points	-2.73 points
Retry the 20% with the lowest head score	-3.0 points	-0.45 points
Retry a random 20% (control)	+1.0 points	-0.67 points
Gated reranking (margin 0)	+1.0 points	+0.24 points
Deeper pass applied to everything	-8.0 points	-12.79 points

The pilot overstated every effect that looked like a gain. With 100 examples, one question is one point, so differences of two to four points are a handful of questions; the full split shows they were noise.

> **WHAT THIS CHANGES IN THE CONCLUSIONS**
> (1) Do not quote the pilot's "81% against 78%": on 9,517 questions head-selected best-of-five is +0.4 points over greedy and not statistically significant. (2) The detection work stands: the head ranks sampled answers far better than the answer-token probability (AUROC 0.883 against 0.815). (3) Turning that ranking into accuracy needs a stronger lever than choosing among five samples or asking the model to look again; the oracle gaps (+4.8 points for best-of-five, +2.9 for retry) say the room exists. (4) The retry and deeper pass should be rerun with a controlled format (enough tokens, final-answer extraction) before drawing conclusions about chain of thought.

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13. Engineering and reproducibility

- **Everything runs on the GT ICE cluster** with one command per stage. `run_qhead_pipeline.sh` runs a dataset end to end (splits, baseline, features, sweeps, analysis), `run_mixture_pipeline.sh` the mixture and swap loss, `run_continual_pipeline.sh` sequential training, `run_corruption_pipeline.sh` the noise loss and its cross-corruption checks, and `run_export_pipeline.sh` saves the trained heads. Each submits Slurm jobs with dependencies.
- **Scripts are checked before long runs.** `check_scripts.sh` validates shell syntax, that every `python -m` module resolves, and (on the cluster) that the scheduler accepts the resource request. Time limits are realistic on purpose, because the cluster caps the total GPU minutes a user can request.
- **36 offline tests** cover the scoring rules, the splits, the unified head, the corruptions, the old and new loss (including a check that the old loss is reproduced exactly when the new terms are off), the grounding metrics, the mixture builder, sequential training, and that the reports build from the real result schema. They need no GPU or network.

- **Feature caching** is what makes the experiments affordable. After one pass of the language model, a pooled sweep (36 or 72 head configurations, three seeds each, on 50,517 training rows) took 3 to 12 minutes per head kind on a single GPU, and a sequential-training run took 1.5 to 3.5 minutes.
- **Layout.** `models/` (model loading and inference), `data/` (dataset acquisition and scoring inputs), `experiments/` (training, analysis, reports), `slurm/`, `results/approach1/` (numbered reports and data), `trained_heads/`, `docs/`, `tests/`.

The design note `docs/approach1_unified_q_head.md` records the dataset choices and the reasons other candidates were rejected.

13.1 Measured run times

Wall-clock times from Slurm accounting, one GPU each. The model passes (features, baselines) dominate the cost; every head experiment on the cached features is short.

Job	Time
Corrupted-image features, ChartQA train (28,299 questions)	82.5 min
Pooled sweep, pre-generation head (36 configurations x 3 seeds)	3.4 min
Pooled sweep, verifier head (36 x 3)	8.0 min
Pooled sweep, unified head (72 x 3)	11.6 min
Sequential training, pre-generation head (4 variants x 3 orders x 3 seeds)	1.5 min
Sequential training, verifier and unified heads	3.4 min each
Export of the 48 pooled-mixture winner heads (retraining)	4.3 min
Original per-dataset pipeline, TextVQA: build / baseline / features	49.7 / 24.6 / 28.4 min
Original per-dataset pipeline, DocVQA: build / baseline / features	67.9 / 39.7 / 62.8 min
Original per-dataset pipeline, ScienceQA: build / baseline / features	2.7 / 8.0 / 11.8 min

14. What it all means

14.1 What is established

- **The detection half works.** A small head on frozen hidden states predicts whether an answer is correct far better than the model's own token probability, on four different kinds of input (AUROC 0.881 against 0.719 pooled), and the ChartQA result reproduces end to end.
- **The signal is cheap.** A linear probe with about 2,000 parameters already captures most of the gain; the largest head has 2.1 million parameters, about 0.1% of the language model.
- **One head is enough.** The unified head does the job of two with no measurable loss, and one head trained on a pool of datasets stays within 0.015 AUROC of heads trained on one dataset.

- **Heads can be extended.** Training on one dataset after another loses 0.011 to 0.046 AUROC depending on the head and the loss (the verifier forgets most), and a 10% replay buffer removes most of that for the verifier.
- **A grounding term is cheap.** It makes the score react to a corrupted image (the share of correct answers whose score falls goes from 0.92 to 0.99 under noise for the unified head) without costing ranking quality, and the effect carries over to corruptions it was not trained on. The pair term, applied alone, costs AUROC.

14.2 What is not established

- **Correction.** Nothing yet shows that acting on a flagged answer improves accuracy. On the full test split (9,517 questions) head-selected best-of-five is +0.4 points over greedy (95% interval -0.04 to +0.84, not significant), prompted retry loses accuracy unless gated (gated gain at most +0.2 points), and the deeper pass loses 12.8 points. The head ranks candidates well, but the ranking is not yet converted into accuracy.
- **Grounding is a proxy.** "The score falls when the image is corrupted" shows the head reacts to the image, not that it tracks the specific evidence for the answer. Noise is an easy corruption.
- **One model size.** Everything is on Qwen3-VL-2B. The 4B check was not run. The exported sequential heads are single seeds.
- **Calibration across datasets.** One temperature is fitted on the pooled validation split. Per-dataset calibration was not studied.

14.3 Next steps

1. Find a stronger lever than choosing among five samples or retrying: the oracles leave +4.8 points (best-of-five) and +2.9 points (retry) on the table, which a better candidate generator or a latent correction could reach.
2. Run a controlled chain-of-thought condition: ask for reasoning, allow enough tokens (the deeper pass used 64), extract the final answer, score it with the head, and compare it with the deeper pass.
3. Build the latent correction step: test-time steering of the hidden state toward a higher head score, evaluated against the prompted baselines above.
4. A harder grounding test: mask only the region the answer comes from; and a softer correctness label for span answers.
5. Check the 4B model, and add extra input features to the head (several layers, image tokens).

14.4 Slide-ready summary

Slide	Message	Number to show
Motivation	Model confidence is saturated and cannot gate	1.0 on 77% to 90% of answers; AUROC 0.719
Method	Small head on frozen hidden states, before and after the answer	2,049 to 2.1M parameters
Result 1	Works on four different datasets	+0.13 to +0.27 AUROC
Result 2	One unified head, one pooled training	AUROC 0.881 pooled

Slide	Message	Number to show
Result 3	New loss makes the score depend on the image	0.92 to 0.99, no AUROC cost
Result 4	Heads can be extended without forgetting	forgetting 0.011 AUROC
Correction	The head ranks candidates well, but selection and retry gain little so far	best-of-five +0.4 points over greedy, not significant (n=9,517); oracle +4.8
Next	Latent correction on the hidden state	not started

Appendix A. Files and references

A.1 Where each result lives

File in results/approach1/	What it contains
01_chartqa_original_study_report.html	The original ChartQA study: heads, losses, the 72-run sweep, test results
02_multidataset_qhead_report.html	Four datasets, the unified head, metric definitions, case studies, bootstrap intervals
03_qhead_comparison_tables.md	Side-by-side metric tables for all scorers and datasets
05_mixture_new_loss_report.html	Pooled mixture, sequential training, the new loss (swap), ablation, grounding
06_corruption_new_loss_report.html	Noise-corrupted version of the new loss, transfer across corruptions, sequential runs
07_correction_experiments_report.html	Prompted retry, reranking, best-of-five, deeper pass (correction experiments)
04_sweep_data/	Machine-readable sweep, sequential and analysis files behind every table
../../trained_heads/README.md	The 540 trained heads and how to load them
../../docs/approach1_unified_q_head.md	Design note: datasets, unified head, loss, protocol

A.2 Glossary

Term	Meaning
Answer-token probability	The geometric mean of the probabilities the model gave its own answer tokens. The untrained baseline.
Linear probe	A trained head with no hidden layer, $s = w^\top h + b$.
Pre-generation head	Scores a question before any answer exists, from the prompt state.
Verifier head	Scores a specific generated answer, from the answer state.
Unified head	One network that reads both states, with a mode that works without the answer.
AUROC	Probability that a random correct answer outranks a random wrong one.
Error AUPRC	How well the wrong answers are packed at the top of a queue sorted by trust.
Order accuracy (grounding)	Share of correct answers whose score falls when the image is corrupted. 0.5 means no effect.
Forgetting	Average drop from the best score a dataset had after its own stage to its score after the last stage.
Replay	Mixing a fixed share of earlier datasets' training rows into each later stage.
Oracle	An unattainable upper bound that uses the ground truth to choose.

A.3 References

1. Guo, C., Pleiss, G., Sun, Y., Weinberger, K. On Calibration of Modern Neural Networks. ICML 2017 (temperature scaling).
2. Ho, J., Jain, A., Abbeel, P. Denoising Diffusion Probabilistic Models. NeurIPS 2020 (the noising step used for the corrupted image).
3. Masry, A., et al. ChartQA: A Benchmark for Question Answering about Charts with Visual and Logical Reasoning. Findings of ACL 2022.
4. Singh, A., et al. Towards VQA Models That Can Read (TextVQA). CVPR 2019.
5. Mathew, M., Karatzas, D., Jawahar, C. V. DocVQA: A Dataset for VQA on Document Images. WACV 2021 (ANLS).
6. Lu, P., et al. Learn to Explain: Multimodal Reasoning via Thought Chains for Science Question Answering (ScienceQA). NeurIPS 2022.
7. Antol, S., et al. VQA: Visual Question Answering. ICCV 2015 (VQA soft accuracy).

A.4 The Excel workbook

Every table in this report and the data behind it is in `approach1_results.xlsx`, one sheet per table. Download it with the button at the top of the report or from `results/approach1/08_approach1_results.xlsx`. The sheets are:

Sheet	What it holds	Rows
Datasets	Split sizes and base accuracy of the Qwen3-VL-2B answers, per dataset and pooled	5
Token baseline	The untrained answer-token probability and an answer-length baseline, test split	5
Original study test	Original ChartQA study, test split, 2,500 questions (from the original report)	5
Original sweep runs	All 72 validation runs of the original ChartQA sweep	72
Head sizes	Exact parameter counts of every head architecture (hidden size 2048)	9
Winners per dataset	Winning configuration of the per-dataset sweeps (four-dataset study)	12
Per-dataset metrics	Test metrics of the five scorers on each dataset (unified rows are means over three seeds)	20
Gate curves	Accuracy of the most trusted fraction kept, and share of all errors captured when the same fraction of least trusted answers is flagged	400
Bootstrap	Paired bootstrap differences (1,000 resamples) with 95% intervals	16
Reliability bins	Calibration bins: mean predicted probability against observed accuracy	66
Case studies	Examples the unified head caught, missed, falsely flagged, or trusted correctly	48
Heads vs linear probes	Best linear probe against best trained head per dataset, head kind and mode, pooled and dataset-alone training, all metrics	32
Mixture swap metrics	Pooled-mixture test metrics of every winner group, swap training corruption, mean and std over three seeds	155
Mixture swap grounding	Share of answers whose score falls under corruption, swap version, pooled and per dataset, with other corruptions where graded	310

Sheet	What it holds	Rows
Mixture swap winners	Winning configuration of every group, swap version	22
Mixture swap sweep	Validation metrics of every configuration swept, swap version (mean over three seeds)	216
Mixture noise metrics	Pooled-mixture test metrics of every winner group, noise training corruption, mean and std over three seeds	155
Mixture noise grounding	Share of answers whose score falls under corruption, noise version, pooled and per dataset, with other corruptions where graded	775
Mixture noise winners	Winning configuration of every group, noise version	22
Mixture noise sweep	Validation metrics of every configuration swept, noise version (mean over three seeds)	216
Ablation 2x2	Pair term and grounding term switched on and off (swap, pooled mixture)	8
Single-dataset losses	Each dataset alone: old and new loss, swap and noise, AUROC and the share of correct answers whose score falls under the training corruption	64
Corruption transfer	Share of correct answers whose score falls under each corruption (swap, noise, mask, shuffle), pooled test split	12
Sequential summary	Final average AUROC, forgetting, backward transfer, retention, and grounding after the last stage, per head, loss and replay	32
Sequential matrices	AUROC and grounding of every dataset after every stage, per order (means over three seeds)	1536
Sequential isolated	Head trained on each dataset alone, scored on that dataset (reference for retention)	128
Trained heads winners	Every exported winner: files, per-seed test AUROC, mean, sweep mean, and the difference	156
Trained heads sequential	Final head of each sequential variant and order (seed 1) and its final test AUROC per dataset	72
Correction retry	Prompted retry policies (100 examples, 25 per dataset)	6
Correction rerank	Scored retry, reranking, gated margins and the oracle (100 examples, 25 per dataset)	12
Correction best-of-five	Selection methods over five sampled candidates, with Wilson 95% intervals	6
Correction deeper pass	Replacing the head-selected candidate with the deeper-pass answer	8
Deeper pass per example	Every deeper-pass answer with its length, whether it hit the 64-token limit, and whether it echoed the prompt	100
Best-of-five candidates	All 500 sampled candidates with head score, answer-token probability and correctness	500
Full best-of-five	Best-of-five selection on the full test split (9,517 questions) against greedy, paired statistics	6
Full retry	Prompted retry, reranking and gated policies on the full test split against no retry	12
Full deeper pass	Deeper pass after best-of-five on the full test split against the selected candidate	8
Full per dataset	Every full-split correction policy by dataset, against its baseline	92
Full deeper pass formats	Deeper-pass answers that hit the 64-token limit or repeated the prompt, and the correct answers they broke	1

Sheet	What it holds	Rows
Timings	Wall-clock minutes from Slurm accounting on the ICE cluster (one GPU each)	8